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Rough solutions for the periodic Korteweg–de Vries equation

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Abstract

The one dimensional Korteweg–de Vries (KdV) equation on a periodic domain and with initial condition in negative Sobolev spaces is studied using ideas from the theory of rough paths. We discuss convergence of Galerkin approximations, a modified Euler scheme and the presence of a random force of white-noise type in time.

In this note we consider the Cauchy problem for the classical Korteweg–de Vries (KdV) equation:

$$\partial_t u(t,\xi) + \partial_\xi^3 u(t,\xi) + \frac{1}{2}\partial_\xi u(t,\xi)^2 = 0, \quad u(0,\xi) = u_0(\xi), \qquad (t,\xi) \in \mathbb{R}\times\mathbb{T} \qquad (1)$$

where the initial condition u_0 belongs to some Sobolev space $H^\alpha(\mathbb{T})$ of the torus $\mathbb{T} = [-\pi,\pi]$. Since the works of Bourgain [2, 14] and Kenig–Ponce–Vega [22, 21, 20], it is known that eq. (1) admits local solutions starting from initial conditions in H^α for any $\alpha > -1/2$. Later Colliander–Keel–Staffilani–Takaoka–Tao [4] proved global well posedness for $\alpha \geq -1/2$. The existence of solutions in negative Sobolev spaces is possible due to the regularizing effect of the dispersive linear term. This regularization is more effective in the whole line and there global solutions exists for any $\alpha > -3/4$ [21, 4]. Other references on the analysis of the KdV equation are [13, 19]. More recently Kappeler–Topalov, taking advantage of the complete integrability of this model, extended the well-posedness to any $\alpha \geq -1$ using the inverse scattering method [18].

Inspired by the theory of *rough paths* we will look for an alternative approach to the construction of (local) solutions of eq. (1).

Rough paths (introduced by Lyons in [25, 24]) allows the study of differential equations driven by irregular functions. They have been applied to the path-wise study of stochastic differential equations driven by Brownian motion, by fractional Brownian motion of any index $H > 1/4$ [6] and other stochastic processes [5, 11]. Part of the theory has been reformulated in terms of the sub-Riemmannian geometry of certain Carnot groups [9]. In [15] we showed how to reinterpret the work of Lyons in the terms of a cochain complex

of finite increments and a related integration theory. The key step is the introduction of a map (called Λ) which encodes a basic fact of rough path theory. We exploited this point of view to treat stochastic partial differential equations of evolution type [23, 17] and to study the initial value problem for a partial differential equation modeling the approximate evolution of random vortex filaments in 3d fluids [1].

We would like to show that the concepts of the theory can be used fruitfully for problems not related to stochastic processes. The periodic Korteweg–de Vries equation is used as a case study for our ideas. The point of view here developed should be applicable also to KdV on the full real line or other dispersive semi-linear equations like the modified KdV or the non-linear Schrödinger equation.

No previous knowledge of rough paths theory is required nor any result on the periodic KdV is used in the following. We took care to make the paper, as much as possible, self-contained. Since the arguments are similar to those used in finite dimensions in [15], the reader can refer to this last paper to gain a wider perspective on the technique and on the stochastic applications of rough paths.

Our method suggests natural ways to numerically approximate solutions of KdV with irregular initial conditions or with random forcing. We describe a numerical scheme for KdV following the lines of the numerical study of SDEs driven by rough paths [7, 12, 10].

The plan of the paper is as follows. In Sect. 1 we start by recasting the KdV eq. (1) in its mild form and to perform some manipulation to motivate the finite-increment equation which we will study in Sect. 2 where we prove existence and uniqueness of local solutions. Then we prove a-priori estimates necessary show that global rough solutions exists for initial conditions in L^2 (Sect. 2.1). In Sect. 2.2 we prove that the rough solutions are limits of Galerkin approximations making rigorous the link between our finite-increment problem and the classical form of the KdV equation. Moreover in Sect. 2.3 we introduce a time discretization scheme and prove its convergence. The equation that we study is just one of a stack of finite-increment equations that can be generated starting from KdV. In Sect. 2.4 we derive the next member of this hierarchy. Finally Sect. 3 addresses the problem of the presence of an additive stochastic forcing.

We collect some longer and technical proofs in the Appendix.

1 Formulation of the problem

The mild form of eq. (1) is

$$u(t,\xi) = U(t)u_0(\xi) + \frac{1}{2}\int_0^t U(t-s)\partial_\xi u(s,\xi)^2\,ds, \qquad t\in[0,T], \xi\in\mathbb{T} \tag{2}$$

where $T>0$ is a fixed horizon and $U(t)$ is the Airy group of isometries of any H^α given by the solution of the linear part of eq. (1). If we denote with $\widehat{\varphi}$ the Fourier coefficients of a function $\varphi:\mathbb{T}\to\mathbb{R}$: $\varphi(\xi)=\sum_{k\in\mathbb{Z}}\widehat{\varphi}(k)e^{ik\xi}$ then the group $U(t)$ acts on φ as $\widehat{U(t)\varphi}(k) = e^{ik^3t}\widehat{\varphi}(k)$. Passing to Fourier variables in eq. (1) and assuming that the initial condition satisfy $\widehat{u}_0(0)=0$ we obtain the integral equation

$$\widehat{u}(t,k) = e^{ik^3t}\widehat{u}_0(k) + \frac{ik}{2}\int_0^t e^{ik^3(t-s)}{\sum_{k_1}}'\widehat{u}(s,k_1)\widehat{u}(s,k-k_1)\,ds, \quad t\in[0,T], k\in\mathbb{Z}_* \tag{3}$$

where $\mathbb{Z}_* = \mathbb{Z}\backslash\{0\}$ and where we use the convention that the primed summation over $k_1 \in \mathbb{Z}$ does not include the terms when $k_1 = 0$ or $k_1 = k$. Indeed if $\widehat{u}_0(0) = 0$ then we will have $\widehat{u}(t,0) = 0$ for any $t \geq 0$.

A last transformation of the original equation consist in using as unknowns the variables $v(t,k) = e^{-ik^3t}\widehat{u}(t,k)$. The integral equations which result is then

$$v(t,k) = v_0(k) + \frac{ik}{2}\sum_{k_1}{}' \int_0^t e^{-i3kk_1k_2s} v(s,k_1)v(s,k_2)\,ds, \quad t \in [0,T], k \in \mathbb{Z}_* \tag{4}$$

where $k_2 = k - k_1$ and $v_0(k) = u_0(k)$. In obtaining this equation we have used the algebraic identity $k^3 - k_1^3 - k_2^3 = 3kk_1k_2$ valid for any triple k, k_1, k_2 such that $k = k_1 + k_2$.

In the rest of the paper we will concentrate on the study of solution to eq. (4) with the fixed point method. Moreover since we will always work in Fourier space we will consider H^α as a space of sequences indexed by $\mathbb{Z}$ with prescribed degree of summability.

Let us fix some notations: we define the spaces $H^\alpha = H^\alpha(\mathbb{T})$ as the completion of the space of functions $\varphi : \mathbb{Z} \to \mathcal{C}$ with a finite number of non-zero values and such that $\varphi(-k) = \overline{\varphi(k)}, k \in \mathbb{Z}$ with respect to the norm $|\varphi|_\alpha^2 := \sum_{k\in\mathbb{Z}}(1+|k|)^{2\alpha}|\varphi(k)|^2$. Moreover we denote H_0^α the subspace of H^α which consists of sequences φ such that $\varphi(0) = 0$. Note that on H_0^α a norm equivalent to $|\cdot|_\alpha$ is given by $\varphi \mapsto (\sum_{k\in\mathbb{Z}_*}|k|^{2\alpha}|\varphi(k)|^2)^{1/2}$. For convenience we will usually use the latter instead that the more usual one. The symbol C in the r.h.s. of estimates denote a positive constant which can be different from line to line.

1.1 Generalized integration

In trying to solve eq. (1) we must face two difficulties: the presence of the k factor and of the convolution, both will come into play when trying to get estimates for the nonlinear term in some H^α space. The only hope we have to regularize both contributions is to exploit the oscillatory character of the phase factor $e^{-i3kk_1k_2s}$. However it is not evident how to bound the integral taking into account the cancellations due to this multiplier.

To gain some insights we would like to perform a Taylor expansion of the eq. (4). Before doing this we take a more general perspective and instead of solving eq. (4) we will try to solve the following problem: find a function v such that $v(0,k) = v_0(k)$ and

$$v(t,k) = v(s,k) + \frac{ik}{2}\sum_{k_1}{}' \int_s^t e^{-i3kk_1k_2\sigma} v(\sigma,k_1)v(\sigma,k_2)\,d\sigma, \quad (t,s) \in [0,T]^2, k \in \mathbb{Z}_*. \tag{5}$$

Of course if we can solve eq. (5) we will have a solution of eq. (4). Moreover eq. (5) encodes the fact that we expect any "piece" of the solution to solve the same integral equation with the right initial condition. Now we can perform an expansion of the r.h.s of eq. (5) by substituting to each occurrence of $v(\sigma,k)$ the corresponding expression given by the integral constraint: a first substitution results in

$$v(t,k) = v(s,k) + \frac{ik}{2}\sum_{k_1}{}' \int_s^t e^{-i3kk_1k_2\sigma} d\sigma v(s,k_1)v(s,k_2) + r(t,s,k) \tag{6}$$

where

$$r(t,s,k) = 2\frac{ik}{2}\sum_{k_1}{}' \int_s^t e^{-i3kk_1k_2\sigma} v(\sigma,k_1)\left(\frac{ik_2}{2}\sum_{k_{21}}{}' \int_s^\sigma e^{-i3k_2k_{21}k_{22}\sigma_1} v(\sigma_1,k_{21})v(\sigma_1,k_{22})\,d\sigma_1\right) d\sigma$$
$$+ \frac{ik}{2}\sum_{k_1}{}' \int_s^t e^{-i3kk_1k_2\sigma}\left(\frac{ik_1}{2}\sum_{k_{11}}{}' \int_s^\sigma e^{-i3k_1k_{11}k_{12}\sigma_1} v(\sigma_1,k_{11})v(\sigma_1,k_{12})\,d\sigma_1\right)$$
$$\times\left(\frac{ik_2}{2}\sum_{k_{21}}{}' \int_s^\sigma e^{-i3k_2k_{21}k_{22}\sigma_2} v(\sigma_2,k_{21})v(\sigma_2,k_{22})\,d\sigma_2\right) d\sigma.$$

This equation shows already an interesting pattern: the expansion allowed us to take out from the σ-integral the unknown at the price of introducing a correction term $r(t,s,k)$ in the form of multiple *iterated integrals*. Let us pretend for a moment that this r term is irrelevant and let us concentrate in the analysis of the remaining term. The oscillatory integral can now be explicitly evaluated:

$$\frac{ik}{2}\int_s^t e^{-i3kk_1k_2\sigma} d\sigma = \frac{e^{-i3kk_1k_2s} - e^{-i3kk_1k_2t}}{6k_1k_2}. \tag{7}$$

The k_i factors at the denominator will give the regularization which we were looking for. Introduce the symmetric bilinear operator $X_{ts} : H^\alpha \times H^\alpha \to H^\alpha$ such that

$$X_{ts}(\varphi_1,\varphi_2)(k) := \sum_{k_1}{}' \frac{e^{-i3kk_1k_2s} - e^{-i3kk_1k_2t}}{6k_1k_2}\varphi_1(k_1)\varphi_2(k_2). \tag{8}$$

An important remark is that the family of operators $\{X_{ts}, (t,s) \in [0,T]^2\}$ satisfy the algebraic equation

$$X_{ts}(\varphi_1,\varphi_2) - X_{tu}(\varphi_1,\varphi_2) - X_{us}(\varphi_1,\varphi_2) = 0, \quad (t,u,s) \in [0,T]^3 \tag{9}$$

as can be easily checked using the definition.

Define the set $\mathcal{D}$ of pairs (γ,α) as $\mathcal{D} = \{(\gamma,\alpha) \in [0,1/2]\times\mathbb{R} : \gamma-\alpha < 1, 3\gamma-\alpha-3/2 < 0\}$. Moreover, given two Banach spaces V, W, denote with $\mathcal{L}(V,W)$ the Banach space of bounded linear operators from V to W endowed with the operator norm.

Lemma 1 *For any couple $(\gamma,\alpha) \in \mathcal{D}$ and for any $\varepsilon \geq 0$ sufficiently small, the operator X_{ts} is bounded from $H^{\alpha-\varepsilon} \times H^{\alpha-\varepsilon}$ to $H^{\alpha+\varepsilon}$ and $|X_{ts}|_{\mathcal{L}((H^{\alpha-\varepsilon})^2, H^{\alpha+\varepsilon})} \leq C_{\gamma,\alpha}|t-s|^\gamma$ where the constant $C_{\gamma,\alpha}$ depends only on α and γ.*

The proof of this lemma is postponed to Appendix A .

Remark 2 *In most of the arguments we take $\varepsilon = 0$. A positive ε will be useful in proving convergence of Galerkin approximations below.*

1.2 Computing the remainder

Eq. (6) has the following abstract form

$$\delta v_{ts} = X_{ts}(v_s, v_s) + r_{ts} \tag{10}$$

where $\delta v_{ts} = v_t - v_s$ and $v_t(k) = v(t,k)$ (the index notation being more comfortable for the following discussion). The first object on the r.h.s. is well understood thanks to Lemma 1, while as we have seen the r term is more complex. What is noteworthy however is that, due to the particular structure of eq. (10) the r term must satisfy a simple algebraic equation. To obtain this equation, note that, for any triple $(t,u,s) \in [0,T]^3$ we have $\delta v_{ts} - \delta v_{tu} - \delta v_{us} = 0$; substituting eq. (10) in this relation we get

$$r_{ts} - r_{tu} - r_{us} = -X_{ts}(v_s, v_s) + X_{tu}(v_u, v_u) + X_{us}(v_s, v_s).$$

Using eq. (9) the expression in the r.h.s. can be simplified in

$$\begin{aligned} r_{ts} - r_{tu} - r_{us} &= -X_{tu}(v_s, v_s) + X_{tu}(v_u, v_u) \\ &= X_{tu}(\delta v_{us}, v_s) + X_{tu}(v_u, \delta v_{us}) \end{aligned} \tag{11}$$

The main observation contained in the work [15] is that sometimes this last linear equation determines r uniquely.

1.3 The Λ-map

To explain the conditions under which we can solve eq. (11) we need some more notation. Given a normed vector space $(V, |\cdot|)$ introduce the vector space $\mathcal{C}_n V \subset C([0,T]^n; V)$ such that $a \in \mathcal{C}_n V$ iff $a_{t_1,\dots,t_n} = 0$ when $t_i = t_j$ for some $1 \le i < j \le n$. We have already introduced the operator $\delta : \mathcal{C}_1 V \to \mathcal{C}_2 V$ defined as $\delta f_{ts} = f_t - f_s$, moreover is useful to introduce another operator $\delta : \mathcal{C}_2 V \to \mathcal{C}_3 V$ defined on continuous functions of *two* parameters on a vector space V as $\delta a_{tus} = a_{ts} - a_{tu} - a_{us}$. The two operators satisfy the relation $\delta\delta f = 0$ for any $f \in \mathcal{C}_1 V$. Moreover if $a \in \mathcal{C}_2 V$ is such that $\delta a = 0$ then there exists $f \in \mathcal{C}_1 V$ such that $a = \delta f$. Denote $\mathcal{Z}\mathcal{C}_3 V = \mathcal{C}_3 V \cap \mathrm{Im}\delta$.

We measure the size of elements in $\mathcal{C}_n V$ for $n = 2, 3$ by Hölder-like norms defined in the following way: for $f \in \mathcal{C}_2 V$ let

$$\|f\|_\mu = \sup_{s,t\in[0,T]} \frac{|f_{ts}|}{|t-s|^\mu}, \quad \text{and} \quad \mathcal{C}_1^\mu V = \{f \in \mathcal{C}_2 V;\ \|f\|_\mu < \infty\}.$$

In the same way, for $h \in \mathcal{C}_3 V$, set

$$\begin{aligned} \|h\|_{\gamma,\rho} &= \sup_{s,u,t\in[0,T]} \frac{|h_{tus}|}{|u-s|^\gamma |t-u|^\rho} \\ \|h\|_\mu &= \inf\left\{ \sum_i \|h_i\|_{\rho_i, \mu-\rho_i};\ h = \sum_i h_i,\ 0 < \rho_i < \mu \right\}, \end{aligned} \tag{12}$$

where the last infimum is taken over all sequences $\{h_i \in \mathcal{C}_3 V\}$ such that $h = \sum_i h_i$ and for all choices of the numbers $\rho_i \in (0, z)$. Then $\|\cdot\|_\mu$ is easily seen to be a norm on $\mathcal{C}_3 V$, and we set $\mathcal{C}_3^\mu V = \{h \in \mathcal{C}_3 V;\ \|h\|_\mu < \infty\}$. Eventually, let $\mathcal{C}_n^{1+} V = \cup_{\mu>1} \mathcal{C}_n^\mu V$ for $n = 2, 3$

and remark that the same kind of norms can be considered on the spaces $\mathcal{ZC}_3 V$, leading to the definition of the spaces $\mathcal{ZC}_3^\mu V$ and $\mathcal{ZC}_3^{1+} V$.

The following proposition is the basic result which allows the solution of equations in the form (11).

Proposition 3 (The Λ-map) *There exists a unique linear map $\Lambda : \mathcal{ZC}_3^{1+} V \to \mathcal{C}_2^{1+} V$ such that*

$$\delta\Lambda = Id_{\mathcal{ZC}_3 V} .$$

Furthermore, for any $\mu > 1$, this map is continuous from $\mathcal{ZC}_3^\mu V$ to $\mathcal{C}_2^\mu V$ and we have

$$\|\Lambda h\|_\mu \leq \frac{1}{2^\mu - 2} \|h\|_\mu, \qquad h \in \mathcal{ZC}_3^{1+} V. \tag{13}$$

This proposition has been first proved in [15]. A simplified proof is contained in [17].

Using the notations just introduced, eq. (11) take the form

$$\delta r_{tus} = X_{tu}(\delta v_{us}, v_s) + X_{tu}(v_u, \delta v_{us}) = [X(\delta v, v) + X(v, \delta v)]_{tus}, \qquad (t,u,s) \in [0,T]^3 \tag{14}$$

where, by construction, the r.h.s. belongs to $\mathcal{ZC}_3 H^\alpha$. If we can prove that it actually belongs to $\mathcal{ZC}_3^z H^\alpha$ for some $z > 1$, Prop. 3 will give us the possibility to state that the unique solution of eq. (14) in $\mathcal{C}_2^{1+} H^\alpha$ is given by $r = \Lambda[X(\delta v, v) + X(v, \delta v)]$.

1.4 Λ equations

Let us come back to our initial problem. Since we aim to work in distributional spaces the rigorous meaning of eq. (5) is not clear (even in a weak sense). By formal manipulations we have been able to recast the initial problem in a finite-difference equation involving the Λ map which reads:

$$\delta v = X(v, v) + \Lambda[X(\delta v, v) + X(v, \delta v)] \tag{15}$$

where we used an abbreviated notation since all the terms have been already described in detail. Note that in this equation the Λ map has replaced the integral so we will call this kind of equations: Λ-*equations*. Instead of solving the integral equation we would like to solve, by fixed-point methods, the Λ-equation (15). Afterwards we will show rigorously that solution to such Λ-equations gives generalized solutions to KdV.

Unfortunately, for the particular form of our X operator, we are not able to show that this equation has solutions. Recall Lemma 1 where we showed that X belongs to $\mathcal{C}_2^\gamma \mathcal{L}(H^\alpha, H^\alpha)$ for $(\gamma, \alpha) \in \mathcal{D}$ so we must expect that $\delta v \in \mathcal{C}_2^\gamma H^\alpha$ since the Λ term will belong (at worst) to $\mathcal{C}_2^{1+} H^\alpha$. But then we have

$$X(\delta v, v) + X(v, \delta v) \in \mathcal{C}_3^{2\gamma} H^\alpha$$

so that we must require $2\gamma > 1$ in order for this term to be in the domain of Λ. The set $\mathcal{D}$ however contains only values of $\gamma \leq 1/2$ so we are not able to study eq. (15) in $\mathcal{C}_1^\gamma H^\alpha$ for $(\gamma, \alpha) \in \mathcal{D}$.

This difficulty can be overcome by pursuiting further the expansion of eq. (5). By looking at eq. (6) we see that the next step in the expansion will generate the following expression

$$\begin{aligned} v(t,k) = v(s,k) &+ \frac{ik}{2}{\sum_{k_1}}' \int_s^t e^{-i3kk_1k_2\sigma} d\sigma\, v(s,k_1)v(s,k_2) \\ &- 2{\sum_{k_1,k_{21}}}' \frac{kk_2}{4} \int_s^t e^{-i3kk_1k_2\sigma} \int_s^\sigma e^{-i3k_2k_{21}k_{22}\sigma_1}\, d\sigma_1\, d\sigma\, v(s,k_1)v(s,k_{21})v(s,k_{22}) \\ &+ r^{(2)}(t,s,k) \end{aligned} \tag{16}$$

where the new remainder $r^{(2)}$ is even more cumbersome that the previous one and here we do not bother to write it down explicitly. In this equation a new (trilinear) operator appear. Define $X^2 \in \mathcal{C}_2\mathcal{L}((H^\alpha)^3, H^\alpha)$ as

$$X^2_{ts}(\varphi_1,\varphi_2,\varphi_3)(k) = -2{\sum_{k_1,k_{21}}}' \frac{kk_2}{4} \int_s^t \int_s^\sigma e^{-i3kk_1k_2\sigma - i3k_2k_{21}k_{22}\sigma_1}\, d\sigma d\sigma_1\, \varphi_1(k_1)\varphi_2(k_{21})\varphi_3(k_{22}) \tag{17}$$

and note that it satisfy the algebraic relation

$$\delta X^2(\varphi_1,\varphi_2,\varphi_3)_{tus} = 2X_{tu}(\varphi_1, X_{us}(\varphi_2,\varphi_3)) \qquad t,u,s \in [0,T] \tag{18}$$

which show that X^2 is related to the previously defined operator X by a "quadratic" relationship which motivates the abuse of the superscript 2 in the definition of X^2.

Remark 4 *Define the unbounded bilinear operator $\dot{X}_s : (H^\alpha)^2 \to H^\alpha$, $s \in [0,T]$ as*

$$\dot{X}_s(\varphi_1,\varphi_2)(k) = \frac{ik}{2}{\sum_{k_1}}' e^{-3kk_1k_2s}\varphi_1(k_1)\varphi_2(k_2).$$

Then for sufficiently regular φ_1, φ_2 we have the representations

$$X_{ts}(\varphi_1,\varphi_2) = \int_s^t \dot{X}_\sigma(\varphi_1,\varphi_2)d\sigma,$$

and

$$X^2_{ts}(\varphi_1,\varphi_{21},\varphi_{22}) = 2\int_s^t \int_s^\sigma \dot{X}_\sigma(\varphi_1, \dot{X}_{\sigma_1}(\varphi_{21},\varphi_{22}))d\sigma d\sigma_1,$$

for any $(t,s) \in [0,T]^2$.

The proof of the following lemma is postponed to Appendix A.

Lemma 5 *There exists unbounded operators $\hat{X}^2 : (H^\alpha)^3 \to H^\alpha, \Phi : (H^\alpha)^2 \to \mathbb{C}$ such that*

$$X^2_{ts}(\varphi_1,\varphi_2,\varphi_3) = \hat{X}^2_{ts}(\varphi_1,\varphi_2,\varphi_3) + \Phi_{ts}(\varphi_1,\varphi_2)\varphi_3 + \Phi_{ts}(\varphi_1,\varphi_3)\varphi_2.$$

Furthermore when $\alpha \geq 1/2$ we have $|\Phi_{ts}|_{\mathcal{L}((H^\alpha)^2,\mathbb{C})} \leq C|t-s|$ and for any couple $(\gamma,\alpha) \in \mathcal{D}$ we have $|\hat{X}^2_{ts}|_{\mathcal{L}((H^\alpha)^3,H^\alpha)} \leq C|t-s|^{2\gamma}$ where the constant C depends only on α and γ. Then $|X^2_{ts}|_{\mathcal{L}((H^\alpha)^3,H^\alpha)} \leq C'_{\gamma,\alpha}|t-s|^{2\gamma}$ for any couple $(\gamma,\alpha) \in \mathcal{D}' = \mathcal{D} \cap \{\alpha \geq 1/2\}$. Call $\mathcal{D}'$ the set of allowed pairs.

Remark 6 *Note that the full operator X^2 can be controlled only in the smaller region $\mathcal{D}'$ of the (γ, α) plane. This limitation has some relation with the low regularity ill-posedness discussed in [3].*

Abstractly we write eq. (16) as

$$\delta v = X(v, v) + X^2(v, v, v) + r^{(2)}. \tag{19}$$

To infer the corresponding Λ-equation we simply apply the δ operator to eq. (19). After some tedious but explicit computations we get

$$\begin{aligned} \delta r^{(2)}_{tus} =& 2X_{tu}(\delta v_{us} - X_{us}(v_s, v_s), v_s) + X_{tu}(\delta v_{us}, \delta v_{us}) \\ &+ X^2_{tu}(\delta v_{us}, v_s, v_s) + X^2_{tu}(v_u, \delta v_{us}, v_s) + X^2_{tu}(v_u, v_u, \delta v_{us}). \end{aligned} \tag{20}$$

2 Rough solutions of KdV

From eq. (20) we can write down a second Λ-equation (beside eq. (15)) associated to the KdV equation:

$$\delta v = X(v, v) + X^2(v, v, v) + v^\flat. \tag{21}$$

with

$$v^\flat = \Lambda[2X(\delta v - X(v, v), v) + X(\delta v, \delta v) + X^2(\delta v, v, v) + X^2(v, \delta v, v) + X^2(v, v, \delta v)].$$

An alternative form of eq. (21) is

$$\delta v = (\mathrm{Id} - \Lambda\delta)[X(v, v) + X^2(v, v, v)] \tag{22}$$

(also eq. (15) can be written in the same form). While more compact this latter form is not very useful for analysis. However it makes evident why the r.h.s. belongs to $\mathrm{Ker}\delta = \mathrm{Im}\delta$.

To give a well-defined meaning to eq. (21) it will be enough that the argument of Λ belongs actually to its domain. Given an allowed pair $(\gamma, \alpha) \in \mathcal{D}'$ which fixes the regularity of X and X^2 sufficient (and natural) requirements for v are

$$\sup_{t\in[0,T]} |v_t|_\alpha < \infty, \qquad \delta v \in \mathcal{C}^\gamma_1 H^\alpha, \qquad \delta v - X(v, v) \in \mathcal{C}^{2\gamma}_2 H^\alpha. \tag{23}$$

Under these conditions $X(\delta v - X(v, v), v) \in \mathcal{C}^{3\gamma}_2 H^\alpha$ and also $X^2(\delta v, v, v) + X^2(v, \delta v, v) + X^2(v, v, \delta v) \in \mathcal{C}^{3\gamma}_2 H^\alpha$. They are in the domain of Λ if $3\gamma > 1$ i.e. $\gamma > 1/3$. This fixes the limiting time regularity for the Λ eq. (21).

Then to define solutions of eq. (21) let us introduce a suitable space to enforce the all the conditions in eq. (23). For any $0 < \eta \leq \gamma$ and consider the complete metric space $\mathcal{Q}_\eta$ whose elements are triples $(y, y', y^\sharp)$ where $y \in \mathcal{C}^\eta_1 H^\alpha$, $y' \in \mathcal{C}^\eta_1 H^\alpha$, $y'_0 = y_0$ and $y^\sharp \in \mathcal{C}^{2\eta}_2 H^\alpha$. Additional requirement is that they satisfy the equation $\delta y = X(y', y') + y^\sharp$. The distance $d_{\mathcal{Q},\eta}$ on $\mathcal{Q}_\eta$ is defined by

$$d_{\mathcal{Q},\eta}(y, z) = |y_0 - z_0| + \|y - z\|_\eta + \|y' - z'\|_\eta + \|y^\sharp - z^\sharp\|_\eta.$$

for any two elements $y, z \in \mathcal{Q}_\eta$.

With abuse of notations we will denote a triple $(y, y', y^\sharp) \in \mathcal{Q}_\eta$ using simply its first component. Moreover we denote with y_0 also the constant path in $\mathcal{Q}_\eta$ with value $(y_0, 0, 0)$.

The main result of the paper is the following theorem.

Theorem 7 *Fix $\gamma \in (1/3, 1/2)$ and $\alpha > 3\gamma - 3/2$. Then, for any $v_0 \in H^\alpha$ the Λ-equation (21) admits a unique solution $v \in \mathcal{Q}_\gamma$ on a small interval $[0, T^*]$ where T^* depends only on the norm of v_0.*

If we write $v = \Theta(X, X^2, v_0)$ for the map

$$\Theta : \mathcal{C}_2^\gamma \mathcal{L}((H^\alpha)^2, H^\alpha) \times \mathcal{C}_2^{2\gamma} \mathcal{L}((H^\alpha)^3, H^\alpha) \times H^\alpha \to \mathcal{Q}_\gamma$$

which send the data to this unique local solution, then Θ is a locally Lipschitz map w.r.t. $(X, X^2, v_0) \in \mathcal{C}_2^\gamma \mathcal{L}((H^\alpha)^2, H^\alpha) \times \mathcal{C}_2^{2\gamma} \mathcal{L}((H^\alpha)^3, H^\alpha) \times H^\alpha$.

The restriction on γ and α ensure that the couple $(\gamma, \alpha) \in \mathcal{D}'$. Then X and X^2 are regular enough so that the proof of the theorem can follow essentially the pattern of a similar result in [15]. The following is an almost self-contained proof.

Proof. Fix $\eta \in (0, \gamma)$ and define a map $\Gamma : \mathcal{Q}_\eta \to \mathcal{Q}_\eta$ by $z = \Gamma(y)$ where

$$z_0 = y_0, \qquad \delta z = X(y, y) + X^2(y, y, y) + z^\flat \tag{24}$$

with

$$z^\flat = \Lambda[2X(y^\sharp, y) + X(\delta y, \delta y) + X^2(\delta y, y, y) + X^2(y, \delta y, y) + X^2(y, y, \delta y)].$$

and we set $z' = y$ and $z^\sharp = X^2(y, y, y) + z^\flat$. For the map Γ to be well defined it is enough that $(\gamma, \alpha) \in \mathcal{D}$ and that $\eta > 1/3$. Indeed if $y \in \mathcal{Q}_\eta$ then all the terms in the argument of the Λ-map belongs to $\mathcal{C}_2^{3\eta} H^\alpha$.

We will proceed in three steps. First we will prove that there exists $T_* \in (0, 1]$ depending only on $|v_0|$ such that for any $T \le T_*$, Γ maps a closed ball B_T of $\mathcal{Q}_\eta$ in itself. Then we will prove that if T is sufficiently small, Γ is actually a contraction on this ball proving the existence of a fixed-point. Finally uniqueness will follow from a standard argument.

Boundedness. The key observation to prove (below) that Γ is a contraction for T sufficiently small is that $\Gamma(y)$ belongs actually to $\mathcal{Q}_\gamma$, indeed $z' = y \in \mathcal{C}_1^\gamma H^\alpha$ and $z^\sharp \in \mathcal{C}_2^{2\gamma} H^\alpha$.

Let us assume that $T \le 1$. First we will prove that

$$d_{\mathcal{Q},\gamma}(\Gamma(y), 0) = d_{\mathcal{Q},\gamma}(z, 0) \le C(1 + d_{\mathcal{Q},\eta}(y, 0))^3. \tag{25}$$

Since $3\eta > 1$ using Prop. 3 we have

$$\begin{aligned} \|z^\flat\|_{3\eta} &\le C\|2X(y^\sharp, y) + X(\delta y, \delta y) + X^2(\delta y, y, y) + X^2(y, \delta y, y) + X^2(y, y, \delta y)\|_{3\eta} \\ &\le C\left[2\|X(y^\sharp, y)\|_{3\eta} + \|X(\delta y, \delta y)\|_{3\eta}\right. \\ &\qquad \left. + \|X^2(\delta y, y, y)\|_{3\eta} + \|X^2(y, \delta y, y)\|_{3\eta} + \|X^2(y, y, \delta y)\|_{3\eta}\right] \end{aligned} \tag{26}$$

The first term is bounded by

$$\|X(y^\sharp, y)\|_{3\eta} \le \|X\|_\eta \|y^\sharp\|_{2\eta} \|y\|_\infty$$

where the Hölder norm $\|X\|_\eta$ is considered with respect to the operator norm on $\mathcal{L}((H^\alpha)^2, H^\alpha)$. Now $\|X\|_\eta \le \|X\|_\gamma$ since we assume $\eta < \gamma$, while $\|y^\sharp\|_{2\eta} \le d_{\mathcal{Q},\eta}(y, 0)$ and finally remark

that $\|y\|_\infty = |y_0| + \|y\|_\eta \leq d_{\mathcal{Q},\eta}(y,0)$ since we assumed $T \leq 1$. So $\|X(y^\sharp, y)\|_{3\eta} \leq C d_{\mathcal{Q},\eta}(y,0)^2$.

Each of the other terms in eq. (26) can be bounded similarly. Then

$$\|z^\flat\|_{3\eta} \leq C(1 + d_{\mathcal{Q},\eta}(y,0))^3 \tag{27}$$

This implies

$$\begin{aligned}\|z^\sharp\|_{2\gamma} &= \|X^2(y,y,y) + z^\flat\|_{2\gamma} \leq \|X^2(y,y,y)\|_{2\gamma} + \|z^\flat\|_{2\gamma} \\ &\leq C\|y\|_\infty^3 + \|z^\flat\|_{2\gamma}\end{aligned}$$

By hypothesis we have $2\gamma < 1 < 3\eta$ so $\|z^\flat\|_{2\gamma} \leq \|z^\flat\|_{3\eta}$ and again we obtain $\|z^\sharp\|_{2\gamma} \leq C(1 + d_{\mathcal{Q},\eta}(y,0))^3$. Finally we have to bound z' and z:

$$\begin{aligned}\|z'\|_\gamma = \|y\|_\gamma &\leq \|X(y',y') + y^\sharp\|_\gamma \leq C\|y'\|_\infty^2 + \|y^\sharp\|_{2\gamma} \\ &\leq C(|y_0|^2 + \|y'\|_\gamma^2) + \|y^\sharp\|_{2\gamma} \leq C(1 + d_{\mathcal{Q},\eta}(y,0))^2\end{aligned}$$

Similarly we can bound z as $\|z\|_\gamma \leq C(1 + d_{\mathcal{Q},\eta}(y,0))^3$. Putting these bounds together we prove eq. (25). Let $\varepsilon = \gamma - \eta > 0$ and observe that

$$\begin{aligned}d_{\mathcal{Q},\eta}(\Gamma(y),0) &= |z_0| + \|z\|_\eta + \|z'\|_\eta + \|z^\sharp\|_{2\eta} \\ &\leq |z_0| + T^\varepsilon \left[\|z\|_\gamma + \|z'\|_\gamma + \|z^\sharp\|_{2\gamma}\right] \\ &\leq |y_0| + T^\varepsilon d_{\mathcal{Q},\gamma}(\Gamma(y),0)\end{aligned}$$

So

$$d_{\mathcal{Q},\eta}(\Gamma(y),0) \leq |y_0| + CT^\varepsilon(1 + d_{\mathcal{Q},\eta}(y,0))^3 \tag{28}$$

Let $B_T = \{y \in \mathcal{Q}_\eta : d_{\mathcal{Q},\eta}(y,0) \leq b_T, y_0 = v_0\}$ be a closed ball of elements in $\mathcal{Q}_\eta$ with initial condition v_0. Let $b_T > 0$ be the solution of the algebraic equation $b_T = |v_0| + CT^\varepsilon(1+b_T)^3$ which exists whenever $T \leq T_*(|v_0|)$ where $T_*(|v_0|)$ is a strictly positive time which depends only on $|v_0|$. If $T \leq T_*(|v_0|)$ and $y \in B_T$ then by eq. (28) we have $d_{\mathcal{Q},\eta}(\Gamma(y),0) \leq |v_0| + CT^\varepsilon(1+b_T)^3 = b_T$ which implies that Γ maps B_T onto itself.

Contraction. By argument similar to those used above we can prove that if $y^{(1)}, y^{(2)}$ are two elements of $\mathcal{Q}_\eta$ then

$$d_{\mathcal{Q},\gamma}(\Gamma(y^{(1)}), \Gamma(y^{(2)})) \leq C d_{\mathcal{Q},\eta}(y^{(1)}, y^{(2)})(1 + d_{\mathcal{Q},\eta}(y^{(1)},0) + d_{\mathcal{Q},\eta}(y^{(2)},0))^2 \tag{29}$$

Then

$$d_{\mathcal{Q},\eta}(\Gamma(y^{(1)}), \Gamma(y^{(2)})) \leq |y_0^{(1)} - y_0^{(2)}| + CT^\varepsilon d_{\mathcal{Q},\eta}(y^{(1)}, y^{(2)})(1 + d_{\mathcal{Q},\eta}(y^{(1)},0) + d_{\mathcal{Q},\eta}(y^{(2)},0))^2 \tag{30}$$

Taking $y_1, y_2 \in B_T$ we have

$$d_{\mathcal{Q},\eta}(\Gamma(y^{(1)}), \Gamma(y^{(2)})) \leq C(1 + 2b_T)^2 T^\varepsilon d_{\mathcal{Q},\eta}(y^{(1)}, y^{(2)}) \tag{31}$$

Let $T_{**} < T_*$ and such that $C(1 + 2b_T)^2 T_{**}^\varepsilon < 1$, eq. (31) implies that Γ is a strict contraction on $B_{T_{**}}$ with a unique fixed-point (in $B_{T_{**}}$).

A standard argument allows to extend the solution to the larger time interval T_*.

Uniqueness. Now assume to have two solutions $v^{(1)}, v^{(2)} \in \mathcal{Q}_\eta$ of eq. (21), i.e. such that $v^{(i)} = \Gamma(v^{(i)})$, $v_0^{(i)} = v_0$ for $i = 1, 2$. Eq. (29) applied to the distance between $\Gamma(v^{(1)})$ and $\Gamma(v^{(2)})$ implies

$$d_{\mathcal{Q},\gamma}(v^{(1)}, v^{(2)}) \leq C d_{\mathcal{Q},\eta}(v^{(1)}, v^{(2)})(1 + d_{\mathcal{Q},\eta}(v^{(1)}, 0) + d_{\mathcal{Q},\eta}(v^{(2)}, 0))^2$$

If we denote $\mathcal{Q}_{\eta,T}$ the space $\mathcal{Q}_\eta$ considered for functions on $[0, T]$ and with $d_{\mathcal{Q},\eta,T}(\cdot,\cdot)$ the corresponding distance, then we have for $S \leq T$ $d_{\mathcal{Q},\eta,S}(y, 0) \leq d_{\mathcal{Q},\eta,T}(y, 0)$ for any $y \in \mathcal{Q}_{\eta,T}$. Then

$$d_{\mathcal{Q},\eta,S}(v^{(1)}, v^{(2)}) \leq C d_{\mathcal{Q},\eta,S}(v^{(1)}, v^{(2)})(1 + d_{\mathcal{Q},\eta,S}(v^{(1)}, 0) + d_{\mathcal{Q},\eta,S}(v^{(2)}, 0))^2$$

and so

$$d_{\mathcal{Q},\eta,S}(v^{(1)}, v^{(2)}) \leq C S^\varepsilon d_{\mathcal{Q},\eta,S}(v^{(1)}, v^{(2)})(1 + d_{\mathcal{Q},\eta,S}(v^{(1)}, 0) + d_{\mathcal{Q},\eta,S}(v^{(2)}, 0))^2$$

so when S is small enough that $CS^\varepsilon(1 + d_{\mathcal{Q},\eta,S}(v^{(1)}, 0) + d_{\mathcal{Q},\eta,S}(v^{(2)}, 0))^2 \leq 1/2$ we must have that $d_{\mathcal{Q},\eta,S}(v^{(1)}, v^{(2)}) = 0$ and then $v_t^{(1)} = v_t^{(2)}$ for any $t \in [0, S]$. By repeating this argument we conclude that the two solution coincide in the whole interval $[0, T]$.

Lipschitz continuity. Modulo some technicalities due to the local character of the solution, the proof of the Lipschitz continuity of the map Θ follows the line of the finite dimensional situation. See [15] for details. ■

2.1 L^2 conservation law and global solutions

It is well known that the KdV equation formally conserves the $L^2 = H^0$ norm of the solution. This conservation law can be used to show existence of global solution when the initial condition is in H^0.

Denote with $\langle \cdot, \cdot \rangle$ the L^2 scalar product: $\langle \varphi, \psi \rangle = \sum_k \overline{\varphi(k)}\psi(k) = \sum_k \varphi(-k)\psi(k)$.

Lemma 8 (L^2 differential conservation law) *For regular $\varphi_1, \varphi_2, \varphi_3$ we have*

$$\langle \varphi_1, \dot{X}_s(\varphi_2, \varphi_3) \rangle + \langle \varphi_2, \dot{X}_s(\varphi_1, \varphi_3) \rangle + \langle \varphi_3, \dot{X}_s(\varphi_2, \varphi_1) \rangle = 0, \qquad s \in [0, T]$$

Proof.

$$\langle \varphi_1, \dot{X}_s(\varphi_2, \varphi_3) \rangle = \sum_{\substack{k,k_1,k_2 \\ k=k_1+k_2}} \frac{ik}{2} e^{-i3kk_1k_2s} \varphi_1(-k)\varphi_2(k_1)\varphi_3(k_2) \tag{32}$$

Make the change of variables $k \to -k$ to get

$$-\sum_{k+k_1+k_2=0} \frac{ik}{2} e^{i3kk_1k_2s} \varphi_1(k)\varphi_2(k_1)\varphi_3(k_2)$$

Now since $k = -k_1 - k_2$ this last quantity is equal to

$$\sum_{k+k_1+k_2=0} \frac{ik_1}{2} e^{i3kk_1k_2s} \varphi_1(k)\varphi_2(k_1)\varphi_3(k_2) + \sum_{k+k_1+k_2=0} \frac{ik_2}{2} e^{i3kk_1k_2s} \varphi_1(k)\varphi_2(k_1)\varphi_3(k_2)$$

If in the first sum we make the change of variables $k_1 \to -k_1$ and in the second $k_2 \to -k_2$ we finally get to $-\langle \varphi_2, \dot{X}_s(\varphi_1, \varphi_3) \rangle - \langle \varphi_3, \dot{X}_s(\varphi_2, \varphi_1) \rangle$, proving the statement. ■

Corollary 9 *For $\varphi \in L^2$,*

$$2\langle \varphi, X^2_{ts}(\varphi,\varphi,\varphi)\rangle + \langle X_{ts}(\varphi,\varphi), X_{ts}(\varphi,\varphi)\rangle = 0.$$

Proof. Note that Lemma 8 implies that $\langle \varphi_1, \dot{X}_s(\varphi_2,\varphi_2)\rangle = -2\langle \varphi_2, \dot{X}_s(\varphi_1,\varphi_2)\rangle$ so we can compute

$$\begin{aligned}
\langle \varphi, X^2_{ts}(\varphi,\varphi,\varphi)\rangle &= 2\int_s^t d\sigma \int_s^\sigma d\sigma_1 \langle \varphi, \dot{X}_\sigma(\varphi, \dot{X}_{\sigma_1}(\varphi,\varphi))\rangle \\
&= -\int_s^t d\sigma \int_s^\sigma d\sigma_1 \langle \dot{X}_{\sigma_1}(\varphi,\varphi), \dot{X}_\sigma(\varphi,\varphi)\rangle \\
&= -\frac{1}{2}\int_s^t d\sigma \int_s^t d\sigma_1 \langle \dot{X}_{\sigma_1}(\varphi,\varphi), \dot{X}_\sigma(\varphi,\varphi)\rangle \\
&= -\frac{1}{2}\langle X_{ts}(\varphi,\varphi), X_{ts}(\varphi,\varphi)\rangle
\end{aligned} \tag{33}$$

which is what we wanted to prove. ■

Theorem 10 (Integral conservation law) *If v is a solution of eq. (21) in $[0,T_*]$ with initial condition $v_0 \in L^2$ then $|v_t|_0^2 = |v_0|_0^2$ for any $t \in [0,T_*]$.*

Proof. We will prove that $\delta\langle v, v\rangle = 0$. Let us compute explicitly this finite increment:

$$[\delta\langle v,v\rangle]_{ts} = \langle v_t, v_t\rangle - \langle v_s, v_s\rangle = 2\langle \delta v_{ts}, v_s\rangle + \langle \delta v_{ts}, \delta v_{ts}\rangle$$

Substituting in this expression the Λ-equation (21) we get

$$\begin{aligned}
[\delta\langle v,v\rangle]_{ts} = {} & 2\langle X_{ts}(v_s,v_s) + X^2_{ts}(v_s,v_s,v_s) + v^\flat_{ts}, v_s\rangle \\
& + \langle X_{ts}(v_s,v_s), X_{ts}(v_s,v_s)\rangle + 2\langle X_{ts}(v_s,v_s), v^\sharp_{ts}\rangle + \langle v^\sharp_{ts}, v^\sharp_{ts}\rangle
\end{aligned}$$

where we set $v^\sharp = X^2(v,v,v) + v^\flat$. Lemma 8 implies that $\langle v_s, X_{ts}(v_s,v_s)\rangle = 0$ while Corollary 9 allows to cancel the X^2 term with the quadratic X term. After the cancellations the increment of the L^2 norm squared is

$$[\delta\langle v,v\rangle]_{ts} = 2\langle v^\flat_{ts}, v_s\rangle + 2\langle X_{ts}(v_s,v_s), v^\sharp_{ts}\rangle + \langle v^\sharp_{ts}, v^\sharp_{ts}\rangle$$

Each term on the r.h.s. belongs at least to $\mathcal{C}^{3\eta}_2\mathbb{R}$ and since $3\eta > 1$ this implies that the function $t \mapsto |v_t|_0^2$ is an Hölder function of index greater than 1 hence it must be constant. ■

Corollary 11 (L^2 global solutions) *If $v_0 \in L^2$ there exist a unique global solutions to the Λ-equation (21).*

Proof. By Thm. 7 there exists a unique local solution up to a time T_* which depends only on $|v_0|_0$. Since $|v_{T_*}|_0 = |v_0|_0$ we can start from T_* and extend uniquely this solution to the interval $[0, 2T_*]$ and then on any interval. ■

2.2 Galerkin approximations

Let P_N the projection on the Fourier modes $|k| \leq N$. Consider the approximate KdV equation

$$\partial_t u^{(N)}(t,\xi) + \partial_\xi^3 u^{(N)}(t,\xi) + \frac{1}{2} P_N \partial_\xi u^{(N)}(t,\xi)^2 = 0,$$
$$u^{(N)}(0,\xi) = u_0^{(N)}(\xi), \qquad (t,\xi) \in \mathbb{R} \times \mathbb{T} \quad (34)$$

where $u_0^{(N)} = P_N u_0$. Is not difficult to show that the unique solution of this equation satisfy the Λ-equation

$$\delta v^{(N)} = (\mathrm{Id} - \Lambda\delta)[X^{(N)}(v^{(N)}, v^{(N)}) + X^{(N),2}(v^{(N)}, v^{(N)}, v^{(N)})] \quad (35)$$

where $v_t^{(N)}(k) = e^{-ik^3 t}\widehat{u^{(N)}}(t,k)$, $X^{(N)} = P_N X(P_N \times P_N)$ and where the operator $X^{(N),2}$ is defined as

$$X_{ts}^{(N),2}(\varphi_1,\varphi_2,\varphi_3) = 2\int_s^t d\sigma \int_s^\sigma d\sigma_1 P_N \dot{X}_\sigma(P_N\varphi_1, P_N\dot{X}_{\sigma_1}(P_N\varphi_2, P_N\varphi_3)) \quad (36)$$

so that

$$\delta X^{(N),2}(\varphi_1,\varphi_2,\varphi_3) = 2X^{(N)}(\varphi_1, X^{(N)}(\varphi_2,\varphi_3)).$$

These are just multi-linear operators in a finite-dimensional space and to have convergence of the Galerkin approximation it would be enough that both converge strongly to their infinite-dimensional analogs X, X^2. Unfortunately it is not difficult to see that $X^{(N),2}$ cannot converge strongly. Indeed a decomposition for $X^{(N),2}$ analogous to that of X^2 described in Lemma 5 holds:

$$X^{(N),2}(\varphi_1,\varphi_2,\varphi_3) = \hat{X}^{(N),2}(\varphi_1,\varphi_2,\varphi_3) + \Phi^{(N)}(\varphi_1,\varphi_2)P_N\varphi_3 + \Phi^{(N)}(\varphi_1,\varphi_3)P_N\varphi_2 \quad (37)$$

where

$$\Phi^{(N)}(\varphi_1,\varphi_2) = \Phi(P_N\varphi_1, P_N\varphi_2).$$

and we will prove in the appendix that

Lemma 12 *Assume $(\gamma,\alpha) \in \mathcal{D}$. Then as $N \to \infty$, $X^{(N)} \to X$ strongly in $\mathcal{C}_2^\gamma\mathcal{L}((H^\alpha)^2, H^\alpha)$ and $\hat{X}^{(N),2} \to X^2$ strongly in $\mathcal{C}_2^\gamma\mathcal{L}((H^\alpha)^3, H^\alpha)$.*

Then we see that the strong convergence of $X^{(N),2}$ would imply that of the maps

$$(\varphi_1,\varphi_2,\varphi_3) \mapsto \Phi(P_N\varphi_1, P_N\varphi_2)P_N\varphi_3 + \Phi(P_N\varphi_1, P_N\varphi_3)P_N\varphi_2$$

and this is clearly not possible.

However the decomposition (37) allows us to write another Λ equation for $v^{(N)}$

$$\delta v^{(N)} = (\mathrm{Id} - \Lambda\delta)[X^{(N)}(v^{(N)}, v^{(N)}) + \hat{X}^{(N),2}(v^{(N)}, v^{(N)}, v^{(N)}) + 2\Phi(v^{(N)}, v^{(N)})v^{(N)}] \quad (38)$$

note that, since $P_N v^{(N)} = v^{(N)}$, in this equation the operator Φ appears at the place of $\Phi^{(N)}$. Of course a similar equation holds for the solution v of the full problem with the couple $X, \hat{X}^2$ instead of $X^{(N)}, \hat{X}^{(N),2}$.

As a consequence, the Lipschitz continuity of the solution of the Λ-equation (38) w.r.t $X, \hat{X}^2$ and v_0 implies the following convergence result.

Corollary 13 *As $N \to \infty$ the Galerkin approximations $v^{(N)}$ converges in $\mathcal{C}_1^\gamma H^\alpha$ to the solution v of the Λ-equation (21) up to a strictly positive time T^* which depends only on the norm of $v_0, X, \hat{X}^2, \Phi$.*

This result is important in that it justifies the identification of solutions to the Λ-equation (15) as generalized solutions of the KdV equation (1).

2.3 A numerical scheme

The solution described by the Λ-equation (22) can be approximated by a discrete Euler-like scheme defined as follows. For any $n > 0$ let $y_0^n = v_0$ and

$$y_i^n = X_{0,1/n}(y_{i-1}^n, y_{i-1}^n) + X^2_{0,1/n}(y_{i-1}^n, y_{i-1}^n, y_{i-1}^n)$$

for $i \geq 1$.

The combination of this scheme with the Galerkin approximation discussed before is an effective numerical approximation scheme for the solutions of KdV with low regularity initial conditions. Indeed the next theorem can be combined with Cor. 13 to obtain effective rates of convergence.

Theorem 14 *Let $\Delta_i^n = y_i^n - y_{i/n}$ and let $T > 0$ be the existence time of the solution described in Thm. 7, then*

$$\sup_{0 \leq i < j \leq nT} \frac{|\Delta_i^n - \Delta_j^n|_\alpha}{|i-j|^\gamma} = O(n^{1-3\gamma}). \tag{39}$$

Proof. Let T be the existence time for the solution v described in theorem 7 and let $N = \lfloor nT \rfloor$. We begin by proving some uniform bounds on the sequence $\{y_i^n, i = 0, \ldots, N\}$. Let

$$q_{ij}^n = X_{i/n,j/n}(y_i^n, y_i^n) + X^2_{i/n,j/n}(y_i^n, y_i^n, y_i^n)$$

so that $y_k^n = \sum_{i=0}^{k-1} q_{i,i+1}$. Given $0 \leq i < j \leq N$ let $\tau_0^1 = i, \tau_1^1 = j$ and define recursively τ_l^k, $k > 0$, $l = 0, \ldots, 2^k$ such that $\tau_{2l}^{k+1} = \tau_l^k$ for $l = 0, \ldots, 2^k$ and $\tau_{2l+1}^{k+1} < \lfloor (\tau_l^k + \tau_{l+1}^k)/2 \rfloor$ for $l = 0, \ldots, 2^k - 1$. Then we have $|\tau_{l+1}^k - \tau_l^k| \leq 1 \vee (j-i)/2^k$ and $y_j^n - y_i^n = \sum_{l=0}^{2^K} q^n_{\tau_l^k, \tau_{l+1}^k}$ where K is such that $(j-i)/2^K \leq 1$. Using the triangular array τ_l^k we rewrite the above expression as a telescopic sum:

$$\begin{aligned} y_j^n - y_i^n &= q_{ij}^n + \sum_{k=1}^{K} \sum_{l=0}^{2^{K-1}} (q^n_{\tau_{2l}^k, \tau_{2l+2}^k} - q^n_{\tau_{2l}^k, \tau_{2l+1}^k} - q^n_{\tau_{2l+1}^k, \tau_{2l+2}^k}) \\ &= q_{ij}^n + \sum_{k=1}^{K} \sum_{l=0}^{2^{k-1}} (\delta q^n)_{\tau_{2l}^k, \tau_{2l+1}^k, \tau_{2l+2}^k} \end{aligned}$$

Up to T, the solution v satisfy the equation $\delta v = X(v,v) + X^2(v,v,v) + v^\flat$ where $\sup_{t \in [0,T]} |v_t|_\alpha + \|v^\flat\|_{3\gamma} \leq C$ so

$$v_{j/n} - v_{i/n} = \sum_{l=i}^{j-1} p_{l,l+1} + \sum_{l=i}^{j-1} v^\flat_{l/n,(l+1)/n}$$

where $p_{ij} = X_{i/n,j/n}(v_{i/n}, v_{i/n}) + X^2_{i/n,j/n}(v_{i/n}, v_{i/n}, v_{i/n})$. Then

$$\Delta^n_j - \Delta^n_i = q^n_{ij} - p_{ij} + \sum_{k=1}^{K} \sum_{l=0}^{2^{k-1}} (\delta q^n - \delta p)_{\tau^k_{2l}, \tau^k_{2l+1}, \tau^k_{2l+2}} - r_{ij}$$

where $r_{ij} = \sum_{l=i}^{j-1} v^\flat_{l/n,(l+1)/n}$. This last term is readily estimated by

$$|r_{ij}| \leq \sum_{l=i}^{j-1} \|v^\flat\|_{3\gamma} n^{-3\gamma} \leq C \left(\frac{j-i}{n}\right) n^{1-3\gamma}$$

uniformly in n. Let

$$M^n_\ell = \sup_{0 \leq i < j \leq \ell} \left(\frac{j-i}{n}\right)^{-1} |\Delta^n_j - \Delta^n_i - q^n_{ij} + p_{ij} + r_{ij}|$$

we want to show that $M^n_N \leq A n^{1-3\gamma}$ uniformly in n for some constant A depending only on the data of the problem: this will imply the statement of the theorem since then

$$|\Delta^n_i| \leq |q^n_{0i} - p_{0i}| + |r_{0i}| + A n^{1-3\gamma} \left(\frac{i}{n}\right) \leq |r_{0i}| + A n^{1-3\gamma} \left(\frac{i}{n}\right) \leq C n^{1-3\gamma}$$

for any $i \leq N$ and

$$|\Delta^n_j - \Delta^n_i| \leq |q^n_{ij} - p^{ij}| + |r_{ij}| + A n^{1-3\gamma} \left(\frac{j-i}{n}\right) \leq C n^{1-3\gamma} \left(\frac{j-i}{n}\right)^\gamma$$

for any $0 \leq i < j \leq N$.

We proceed by induction on ℓ. For $\ell = 1$ the statement is clearly true since $\Delta^n_1 - \Delta^n_0 - q^n_{0,1} + p_{0,1} + r_{0,1} = 0$, moreover for the same reason we have, for all l that $\Delta^n_{l+1} - \Delta^n_l - q^n_{l,l+1} + p_{l,l+1} + r_{l,l+1} = 0$. Assume then that $M^n_{\ell-1} \leq A$ for some $\ell > 0$. The basic observation is that when $|i-j| \leq \ell$ the sums

$$\Delta^n_j - \Delta^n_i - q^n_{ij} + p_{ij} + r_{ij} = \sum_{k=1}^{K} \sum_{l=0}^{2^{k-1}} (\delta q^n - \delta p)_{\tau^k_{2l}, \tau^k_{2l+1}, \tau^k_{2l+2}}$$

can be estimated in terms of $M^n_{\ell-1}$ and various norms of v,X,X^2 much like in the proof of Thm. 7. The bound has the form

$$|(\delta q^n - \delta p)_{\tau^k_{2l}, \tau^k_{2l+1}, \tau^k_{2l+2}}| \leq C \left(\frac{j-i}{2^k n}\right)^{3\gamma} (1 + M^n_{\ell-1})^3 (M^n_{\ell-1} + n^{1-3\gamma})$$

where $C = C(v_0, X, X^2)$ and where the factor $n^{1-3\gamma}$ is due to the previous estimate on r_{ij}. Then since $3\gamma > 1$ and

$$\sum_{k=1}^{K} \sum_{l=0}^{2^{k-1}} \left(\frac{j-i}{2^k n}\right)^{3\gamma} \leq \left(\frac{j-i}{n}\right)^{3\gamma} \sum_{k=1}^{\infty} 2^{k(1-3\gamma)} \leq C \left(\frac{j-i}{n}\right)^{3\gamma} \leq C \left(\frac{\ell}{n}\right)^{3\gamma-1} \left(\frac{j-i}{n}\right)$$

we get

$$M_\ell^n \leq C(1+M_{\ell-1}^n)^4 \sum_{k=1}^{K}\sum_{l=0}^{2^{k-1}} \left(\frac{j-i}{2^k n}\right)^{3\gamma} \leq C(1+M_{\ell-1}^n)^3(M_{\ell-1}^n + n^{1-3\gamma}) \left(\frac{\ell}{n}\right)^{3\gamma-1}$$

Let $m_\ell^n = n^{3\gamma-1}M_\ell^n$, then, for n large enough $m_\ell^n \leq F(m_{\ell-1}^n)$ where F is the increasing map

$$\mathbb{R}_+ \ni m \mapsto F(m) = C(1+m)^4 \left(\frac{\ell}{n}\right)^{3\gamma-1} \in \mathbb{R}_+$$

which, for ℓ/n small enough has a unique attracting fix-point under iteration starting from 0. In particular the iterations stay bounded and if we set $x_{i+1} = F(x_i)$, $x_0 = 0$ we have $A = \sup_i F(x_i) < \infty$ and $m_\ell^n \leq x_\ell \leq A$. By repeating this argument it is easy to prove that the bound holds for all $\ell \leq nT$, i.e. in the whole existence interval found in Thm. 7. ■

Remark 15 *With a bit more of work it is possible to prove the existence of the solution stated in Thm. 7 using directly the discrete approximation as done by Davie [7] for rough differential equations.*

2.4 Higher order Λ equations

Further expansion of eq. (16) generate a hierarchy of Λ-equations for KdV. The next one is given by

$$\delta v = (\mathrm{Id} - \Lambda\delta)[X(v^{\times 2}) + X^2(v^{\times 3}) + X^{3a}(v^{\times 4}) + X^{3b}(v^{\times 4})] \tag{40}$$

where $v^{\times n} = (v, \ldots, v)$ (n times) and where X^{3a}, X^{3b} are operators increments respectively defined as

$$X_{ts}^{3a}(\varphi_1,\varphi_2,\varphi_3,\varphi_4) = \int_s^t d\sigma_1 \dot{X}_{\sigma_1}(\varphi_1, \int_s^{\sigma_1} d\sigma_2 \dot{X}_{\sigma_2}(\varphi_2, \int_s^{\sigma_2} d\sigma_3 \dot{X}_{\sigma_3}(\varphi_3,\varphi_4))) \tag{41}$$

and

$$X_{ts}^{3b}(\varphi_1,\varphi_2,\varphi_3,\varphi_4) = \int_s^t d\sigma_1 \dot{X}_{\sigma_1}(\int_s^{\sigma_1} d\sigma_2 \dot{X}_{\sigma_2}(\varphi_1,\varphi_2), \int_s^{\sigma_1} d\sigma_3 \dot{X}_{\sigma_3}(\varphi_3,\varphi_4)) \tag{42}$$

which satisfy the following relations with X and X^2:

$$\delta X^{3a}(\varphi_1,\varphi_2,\varphi_3,\varphi_4) = X(\varphi_1, X^2(\varphi_2,\varphi_3,\varphi_4)) + X^2(\varphi_1,\varphi_2,X(\varphi_3,\varphi_4)) \tag{43}$$

and

$$\begin{aligned}\delta X^{3b}(\varphi_1,\varphi_2,\varphi_3,\varphi_4) = {} & X(X(\varphi_1,\varphi_2),X(\varphi_3,\varphi_4)) + X^2(\varphi_3,\varphi_4,X(\varphi_1,\varphi_2)) \\ & + X^2(\varphi_1,\varphi_2,X(\varphi_3,\varphi_4))\end{aligned} \tag{44}$$

As we report elsewhere [16] the Hopf algebra of rooted trees is the natural language to describe this hierarchy of equations and the algebraic relations between the various operators. In this special case however these relations can be easily checked by direct computations.

Using Lemmas 1 and 5 we can show that the r.h.s. of the eqns. (43) and (44) belongs to the domain of Λ and so the equations can be used to express X^{3a}, X^{3b} in function of X, X^2 and prove that

Corollary 16 *For $(\gamma,\alpha) \in \mathcal{D}'$ with $\gamma > 1/3$ we have $X^{3a}, X^{3b} \in \mathcal{C}_2^\gamma\mathcal{L}((H^\alpha)^4, H^\alpha)$.*

3 Additive stochastic forcing

As another application of this approach we would like to discuss the presence of an additive random force in the KdV eq. (1):

$$\partial_t u(t,\xi) + \partial_\xi^3 u(t,\xi) + \frac{1}{2}\partial_\xi u(t,\xi)^2 = \Phi \partial_t \partial_\xi B(t,\xi) \tag{45}$$

where Φ is a bounded linear operator on H^0 and $\partial_t\partial_\xi B$ a white noise on $\mathbb{R}\times\mathbb{T}$. We assume that Φ is diagonal in Fourier space: $\Phi e_k = \lambda_k e_k$ where $\{e_k\}_{k\in\mathbb{Z}}$ is the orthonormal basis $e_k(\xi) = e^{ik\xi}/\sqrt{2\pi}$ and that $\lambda_0 = 0$. In this way the noise does not affect the zero mode.

For the rest of this section fix and allowed pair $(\gamma,\alpha)\in\mathcal{D}'$ such that $\gamma\in(1/3,1/2)$ and assume that

$$\sum_{q\in\mathbb{Z}_*} |q|^{2\alpha}\lambda_q^2 < \infty. \tag{46}$$

which implies that Φ is Hilbert-Schmidt from L^2 to H^α.

The transformed integral equation (analogous to eq. (4)) associated to (45) is

$$v(t,k) = v_0(k) + \lambda_k \beta_t^k + \int_0^t \dot{X}_s(v(s),v(s))(k)\,ds, \quad t\in[0,T], k\in\mathbb{Z}_* \tag{47}$$

where $\{\beta_\cdot^k\}_{k\in\mathbb{Z}_*}$ is a family of complex-valued, zero mean, Brownian motions such that $\beta^{-k} = \overline{\beta^k}$ and with covariance $\mathbb{E}[\overline{\beta}_t^k \beta_s^q] = \delta_{k,q}(t\wedge s)$. The relation between the initial noise B and the family $\{\beta_\cdot^k\}_{k\in\mathbb{Z}_*}$ is given by $\beta_t^k = \langle e_k, \int_0^t U(-s)\partial_\xi\partial_s B(s,\cdot)ds\rangle_{H^0}$.

Eq. (47) can be expanded in the same way as we have done before and the first interesting Λ-equation which appears is the following:

$$\delta v = (\mathrm{Id} - \Lambda\delta)[X(v,v) + \delta w + X^2(v,v,v) + X^w(v)]. \tag{48}$$

Here $\delta w(k)_{ts} = \lambda_k(\beta_t^k - \beta_s^k)$ and the random operator $X_{ts}^w : H^\alpha \to H^\alpha$ is given by

$$X_{ts}^w(\varphi) = \int_s^t d\sigma \dot{X}_\sigma(\varphi, \delta w_{\sigma s}) \tag{49}$$

and satisfy the equation

$$\delta X_{tus}^w = X_{tu}(\varphi, \delta w_{us}). \tag{50}$$

From condition (46) we have that w is a.s. a Brownian motion with values in H^α and then by classical results that $\delta w \in \mathcal{C}_2^\rho H^\alpha$ for any $\rho < 1/2$ and a-fortiori $\delta w\in\mathcal{C}_2^\gamma H^\alpha$ by choosing $\rho\in[\gamma,1/2)$.

To prove that a sufficiently regular version of the Gaussian stochastic process X^w exists we will use a generalization of the classic Garsia-Rodemich-Rumsey lemma which has been proved in [15].

Lemma 17 *For any $\theta > 0$ and $p\geq 1$, there exists a constant C such that for any $R\in\mathcal{C}_2 V$ ($(V,|\cdot|)$ some Banach space), we have*

$$\|R\|_\theta \leq C(U_{\theta+2/p,p}(R) + \|\delta R\|_\theta), \tag{51}$$

where

$$U_{\theta,p}(R) = \left[\int_{[0,T]^2} \left(\frac{|R_{ts}|}{|t-s|^\theta}\right)^p dtds\right]^{1/p}.$$

We would like that X^w behaves not worser than X^2. This is indeed the case as stated in the next lemma.

Lemma 18 *Under condition (46) we have $X^w \in \mathcal{C}_2^{2\gamma}\mathcal{L}(H^\alpha, H^\alpha)$ a.s..*

Proof. After an integration by parts, X^w can be rewritten as

$$X^w_{ts}(\varphi) = X_{ts}(\varphi, \delta w_{ts}) - \int_s^t X_{\sigma s}(\varphi, dw_\sigma) = X_{ts}(\varphi, \delta w_{ts}) + I_{ts} \tag{52}$$

The first term in the r.h.s. belongs to $\mathcal{C}_2^{2\gamma}\mathcal{L}(H^\alpha, H^\alpha)$ path-wise:

$$|X_{ts}(\varphi, \delta w_{ts})|_\alpha \leq |X_{ts}|_{\mathcal{L}(H^\alpha \times H^\alpha, H^\alpha)} |\delta w_{ts}|_\alpha |\varphi|_\alpha \leq \|X\|_\gamma \|\delta w\|_\gamma |\varphi|_\alpha |t-s|^{2\gamma}.$$

Let us estimate the Wiener integral I_{ts}. Take $\gamma_1 < \gamma$ so that (γ_1, α) is still an allowed pair,

$$\begin{aligned}
\mathbb{E}\,|I_{ts}|^2_\alpha &= \sum_q \lambda_q^2 \int_s^t |X_{\sigma s}(e_q, \varphi)|^2_\alpha d\sigma \\
&\leq C \sum_q \lambda_q^2 \int_s^t |\sigma - s|^{2\gamma_1} d\sigma \sum_k |k|^{2\alpha} \left| \sum_{k_1} |k|^{\gamma_1} |k_1|^{\gamma_1 - 1} |k_2|^{\gamma_1 - \alpha - 1} e_q(k_1) \psi(k_2) \right|^2 \\
&\leq C \sum_q \lambda_q^2 \int_s^t |\sigma - s|^{2\gamma_1} d\sigma \sum_k |k|^{2\alpha + 2\gamma_1} |q|^{2\gamma_1 - 2} |k-q|^{2\gamma_1 - 2\alpha - 2} |\psi(k-q)|^2
\end{aligned}$$

where we used a bound for the kernel of X which has been introduced in the proof of Lemma 1 and where $\psi(k) = |k|^\alpha \varphi(k)$. The function $k \mapsto |k|^{2\alpha+2\gamma_1}|k-q|^{2\gamma_1-2\alpha-2}$ is bounded by $C|q|^{2\alpha+2\gamma_1}$ so

$$\mathbb{E}\,|I_{ts}|^2_\alpha \leq C \sum_q \lambda_q^2 |q|^{4\gamma_1 + 2\alpha - 2} |t-s|^{1+2\gamma_1} |\psi|_0^2$$

Choosing $\gamma_1 = 2\gamma - 1/2 + \varepsilon/4 \leq \gamma$ (with $\varepsilon > 0$ small enough) we obtain

$$\mathbb{E}\,|I_{ts}|^2_\alpha \leq C \sum_q \lambda_q^2 |q|^{8\gamma + 2\alpha - 4 + \varepsilon} |t-s|^{4\gamma + \varepsilon/2} |\varphi|^2_\alpha \leq C |t-s|^{4\gamma+\varepsilon/2} |\varphi|^2_\alpha$$

The q sum is bounded since we can choose $\varepsilon > 0$ small enough so that $8\gamma - 4 + \varepsilon \leq 0$ which allows us to bound $|q|^{8\gamma+2\alpha-4+\varepsilon} \leq |q|^{2\alpha}$ and then conclude using condition (46) on the summability of λ_q.

Since the Wiener integral I_{ts} is a Gaussian r.v. the L^2 bound is enough to obtain an L^p bound for any $p \geq 2$:

$$\mathbb{E}\,|I_{ts}|^p_\alpha \leq C_p |\varphi|^p_\alpha |t-s|^{2p\gamma + p\varepsilon/4}$$

From this bound we deduce that, choosing p sufficiently large, we have $U_{2\gamma+2/p,p}(I) < \infty$ a.s.. Moreover $\delta I_{tus}(\varphi) = X_{tu}(\varphi, \delta w_{us})$ so that δI can be bounded path-wise in $\mathcal{C}_3^{2\gamma}\mathcal{L}(H^\alpha, H^\alpha)$. Then Lemma 17 implies that $\|I\|_{2\gamma} \leq C(U_{2\gamma+2/p,p}(I) + \|\delta I\|_{2\gamma}) < \infty$ a.s. ending the proof. ■

Modifying a bit the proof of Thm. 7 is not difficult to prove the following.

Theorem 19 *Eq. (48) has a unique local solution in $\mathcal{C}^\gamma H^\alpha$ for any initial condition in H^α.*

Since we can take γ arbitrarily close to $1/3$ we obtain solutions for noises with values in H^α for any $\alpha > -1/2$. In this way we essentially cover the result of De Bouard-Debussche-Tsutsumi [8]. Their approach consist in modifying Bourgain's method to handle Besov spaces in order to compensate for the insufficient Sobolev time regularity of Brownian motion.

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A Regularity of X and X^2

Some elementary calculus results needed in the proofs of this appendix are the subject of the next lemma.

Lemma 20 *If $\zeta > 1$*

$$\sum_{n\in\mathbb{Z}\setminus\{0,m\}} |n|^{-\zeta}|n-m|^{-\zeta} \le C_\zeta |m|^{-\zeta}$$

and for any $\theta < 2\zeta - 1$

$$\sum_{n\in\mathbb{Z}\setminus\{0,\pm m\}} |n+m|^{-\zeta}|n|^{\theta}|n-m|^{-\zeta} \le C_{\zeta,\theta} |m|^{\theta-\zeta}$$

A.1 Proof of Lemma 1

. We must prove that, for $(\gamma,\alpha)\in\mathcal{D}$ there exists $\varepsilon > 0$ such that

$$\left[\sum_{k\in\mathbb{Z}_*} |k|^{2\alpha+2\varepsilon}\left|{\sum_{k_1}}' \frac{e^{-i3kk_1k_2s} - e^{-i3kk_1k_2t}}{6k_1k_2}\varphi_1(k_1)\varphi_2(k_2)\right|^2\right]^{1/2} \le C|t-s|^\gamma |\varphi_1|_{\alpha-\varepsilon}|\varphi_2|_{\alpha-\varepsilon}$$

To begin we bound

$$\left|\frac{e^{-i3kk_1k_2s} - e^{-i3kk_1k_2t}}{6k_1k_2}\right| \le C|t-s|^\gamma |k|^\gamma |k_1|^{\gamma-1}|k_2|^{\gamma-1}$$

for any $\gamma\in[0,1]$. We use the conventions that, if $\beta < 0$ then $|0|^\beta = 0$ (with integer argument). The constant in this bound is independent on anything else. By duality it will be enough to prove that, for any $\rho,\psi_1,\psi_2\in H^0_0$ we have

$$\sum_{k,k_1} |k|^{\alpha+\varepsilon+\gamma}|k_1k_2|^{\gamma-\alpha+\varepsilon-1}|\rho(k)||\psi_1(k_1)||\psi_2(k_2)| \le C|\rho|_0|\psi_1|_0|\psi_2|_0. \tag{53}$$

where we understand that summations are done over $\mathbb{Z}$ excluding values for which $k_1 = 0$ or $k = 0$ or $k_2 = 0$.

To prove eq. (53) for any couple $(\gamma, \alpha) \in \mathcal{D}$ we proceed in three steps: we will prove first that it holds for (γ, α) in two regions $\mathcal{D}_1, \mathcal{D}_2 \subset \mathbb{R}^2$ and then by interpolation we will extend its validity to the convex hull of $\mathcal{D}_1 \cup \mathcal{D}_2$ which will result to be exactly $\mathcal{D}$.

First region. Let $\mathcal{D}_1 = \{(\gamma, \alpha) : \gamma - \alpha < 1/2, \gamma \in [0, 1/2)\}$ and in this paragraph fix $(\gamma, \alpha) \in \mathcal{D}_1$. Applying Cauchy-Schwarz inequality w.r.t the k_1 summation on the l.h.s. of eq. (53) we obtain

$$\sum_k |k|^{\alpha+\gamma+\varepsilon} |\rho(k)| \left[\sum_{k_1} |k_1 k_2|^{2\gamma - 2\alpha + 2\varepsilon - 2} \right]^{1/2} \left[\sum_{k_1} |\psi_1(k_1)|^2 |\psi_2(k_2)|^2 \right]^{1/2} \tag{54}$$

Observe that, since $2\gamma - 2\alpha - 1 < 0$, Lemma 20 allows us to bound the sum over k_1 as

$$\sum_{k_1} |k_1 k_2|^{2\gamma - 2\alpha + 2\varepsilon - 2} \leq C |k|^{2\gamma - 2\alpha + 2\varepsilon - 2}$$

for sufficiently small $\varepsilon > 0$. Injecting this bound in eq. (54) we find

$$C \sum_k |k|^{2\gamma - 1 + 4\varepsilon} |\rho(k)| \left[\sum_{k_1} |\psi_1(k_1)|^2 |\psi_2(k_2)|^2 \right]^{1/2} \tag{55}$$

Given that $\gamma < 1/2$ we have $|k|^{2\gamma - 1 + 4\varepsilon} \leq 1$ if we choose ε small enough. Taking this into account in the last bound and applying Cauchy-Schwarz w.r.t to the k summation we can bound our initial expression by $C|\rho|_0 [\sum_{k,k_1} |\psi_1(k_1)|^2 |\psi_2(k_2)|^2]^{1/2}$. But now observe that a change of variables is enough to estimate this last double summation:

$$\sum_{k,k_1} |\psi_1(k_1)|^2 |\psi_2(k_2)|^2 = \sum_{k_2 k_1} |\psi_1(k_1)|^2 |\psi_2(k_2)|^2 = |\psi_1|_0^2 |\psi_2|_0^2$$

so that we have proven eq. (53) in the case of $(\gamma, \alpha) \in \mathcal{D}_1$.

Second region. Let $\mathcal{D}_2 = \{(\gamma, \alpha) \in \mathbb{R}^2 : \gamma \in [0, 1/2), \gamma - \alpha < 1, \gamma + \alpha < -1/2\}$ and fix $(\gamma, \alpha) \in \mathcal{D}_2$. Apply Cauchy-Schwartz to the l.h.s. of eq. (53) to obtain

$$\left[\sum_k |k|^{2\gamma + 2\alpha + 2\varepsilon} \right]^{1/2} \left[\sum_k |\rho(k)|^2 \left(\sum_{k_1} |k_1 k_2|^{\gamma - \alpha - 1 + 2\varepsilon} |\psi_1(k_1)| |\psi_2(k_2)| \right)^2 \right]^{1/2}$$

Choose ε small enough so that $\gamma + \alpha + \varepsilon < -1/2$ and $\gamma - \alpha + \varepsilon \leq 1$. Then these two inequalities imply that the first sum over k is bounded and that we can discard the powers of k_1 and k_2 in the second factor. Finally we get

$$C \left[\sum_k |\rho(k)|^2 \left(\sum_{k_1} |\psi_1(k_1)| |\psi_2(k_2)| \right)^2 \right]^{1/2}$$

Now observe that $\sum_{k_1} |\psi_1(k_1)| |\psi_2(k_2)| \leq |\psi_1|_0 |\psi_2|_0$ and then we can bound the last display by $|\rho|_0 |\psi_1|_0 |\psi_2|_0$, proving eq. (53) in the region $\mathcal{D}_2$.

Interpolation. Let $(\gamma_1, \alpha_1) \in \mathcal{D}_1 \cup \mathcal{D}_2$ and $(\gamma_2, \alpha_2) \in \mathcal{D}_1 \cup \mathcal{D}_2$, consider for $t \in [0,1]$, $\gamma(t) = t\gamma_1 + (1-t)\gamma_2$, $\alpha(t) = t\alpha_1 + (1-t)\alpha_2$. Moreover let ε be the smaller of the two ε related to the couple (γ_1, α_1) and (γ_2, α_2). Let

$$A(\gamma, \alpha) = \sum_{k,k_1} |k|^{\alpha+\varepsilon+\gamma} |k_1|^{\gamma-\alpha+\varepsilon-1} |k_2|^{\gamma-\alpha+\varepsilon-1} |\rho(k)||\psi_1(k_1)||\psi_2(k_2)|.$$

By using Hölder inequality with conjugate exponents $p = 1/t$ and $q = 1/(1-t)$ we have

$$A(\gamma(t), \alpha(t)) \leq A(\gamma_1, \alpha_1)^{1/p} A(\gamma_2, \alpha_2)^{1/q}$$

so since we have $A(\gamma_i, \alpha_i) \leq C_{\gamma_i, \alpha_i} |\psi_1|_0 |\psi_2|_0 |\rho|_0$ for $i = 1, 2$, we obtain

$$A(\gamma(t), \alpha(t)) \leq C^t_{\gamma_1, \alpha_1} C^{1-t}_{\gamma_2, \alpha_2} |\psi_1|_0 |\psi_2|_0 |\rho|_0$$

thus proving eq. (53) for the couple $(\gamma(t), \alpha(t))$ for any $t \in [0,1]$. This implies that the bound (53) holds for any couple (γ, α) in the convex hull of the set $\mathcal{D}_1 \cup \mathcal{D}_2$. It is not difficult to see that this convex hull corresponds to the region $\mathcal{D}$ of the statement. This conclude the proof. ■

A.2 Proof of Lemma 5

. We start the argument as in the proof of Lemma 1. Let $\Psi_{ts} = X_{ts}(\varphi_1, \varphi_{21}, \varphi_{22})$, then

$$\Psi_{ts}(k) = -2 \sum_{k_1, k_{21}}{}' \frac{k k_2}{4} \int_s^t \int_s^\sigma e^{-i3kk_1k_2\sigma - i3k_2k_{21}k_{22}\sigma_1} \, d\sigma d\sigma_1 \, \varphi_1(k_1)\varphi_{21}(k_{21})\varphi_{22}(k_{22}).$$

Note that it is enough to restrict the sums over the set of k_i for which $kk_1k_2 \neq 0$ and $k_2k_{21}k_{22} \neq 0$. Then the multiplier

$$M = -\frac{k k_2}{2} \int_s^t \int_s^\sigma e^{-i3kk_1k_2\sigma - i3k_2k_{21}k_{22}\sigma_1} \, d\sigma d\sigma_1; \tag{56}$$

has two different behaviors depending on the quantity $h = kk_1k_2 + k_2k_{21}k_{22}$ being zero or not. Split Ψ accordingly and let

$$\Psi^1_{ts}(k) = -2 \sum_{k_1, k_{21} : h=0}{}' M \, \varphi_1(k_1)\varphi_{21}(k_{21})\varphi_{22}(k_{22})$$

and $\Psi^2 = \Psi - \Psi^1$. We have $h = k_2(k_1 + k_{21})(k - k_{21})$ so in the sum for Ψ^1 the only relevant contributions came from the case where $k_1 + k_{21} = 0$ or $k - k_{21} = 0$. In the first case we have $k_{22} = k$ while in the second $k_{22} = -k_1$ so defining

$$\Phi_{ts}(\varphi_1, \varphi_{21})(k) = -2 \sum_{k_1}{}' M|_{k_{21}=-k_1, k_{22}=k} \, \varphi_1(k_1)\varphi_{21}(-k_1)$$

we have $\Psi_{ts} = \Psi^2_{ts} + \Phi_{ts}(\varphi_1, \varphi_{21})\varphi_{22} + \Phi_{ts}(\varphi_1, \varphi_{22})\varphi_{21}$.

In Lemma 21 below we prove that, if $\gamma \leq 1/2$ and $h \neq 0$ then $|M| \leq C(A_1 + A_2)|t-s|^{2\gamma}$ where

$$A_1 = |k||k_2||kk_1k_2|^{\gamma-1}|k_2k_{21}k_{22}|^{\gamma-1}$$

and

$$A_2 = |k|^{1-\gamma}|k_2|^{2\gamma-1}|k_1|^{-\gamma}|k-k_{21}|^{2\gamma-1}|k_1+k_{21}|^{2\gamma-1}|k_{21}|^{\gamma-1}|k_{22}|^{\gamma-1}$$

while if $h=0$, $|M| \leq C|k_1|^{-1}|t-s|$.

Bound for Φ. We have $|\Phi(\varphi_1,\varphi_{21})| \leq C|t-s|\Xi_0$ where

$$\Xi_0 = \sum_{k_1} |k_1|^{-2\alpha-1}|\rho(k)||\psi_1(k_1)||\psi_{21}(-k_1)|$$

It is clear that these expressions can be controlled with the L^2 norms of the test functions as soon as $-2\alpha-1 \leq 0$.

Bound for Ψ^2. In order to bound Ψ^2 it will be enough to bound separately the forms

$$\Xi_1 = \sum_{k,k_1,k_{21}} A_1|k|^{\alpha}|k_1|^{-\alpha}|k_{21}|^{-\alpha}|k_{22}|^{-\alpha}|\rho(k)||\psi_1(k_1)||\psi_{21}(k_{21})||\psi_{22}(k_{22})|$$

and

$$\Xi_2 = \sum_{k,k_1,k_{21}} A_2|k|^{\alpha}|k_1|^{-\alpha}|k_{21}|^{-\alpha}|k_{22}|^{-\alpha}|\rho(k)||\psi_1(k_1)||\psi_{21}(k_{21})||\psi_{22}(k_{22})|$$

in terms of the L^2 norms of the functions $\rho, \psi_1, \psi_{21}, \psi_{22}$.

The expression A_1 is nicely factorized in its dependence on the couple k_{21}, k_{22} and the form Ξ_1 can be easily bounded using (twice and iteratively) the same arguments as in Lemma 1. We will concentrate on the form Ξ_2 which require different estimates.

Write Ξ_2 as

$$\Xi_2 = \sum_{k,k_1,k_{21}} |k|^{1-\gamma+\alpha} A_3|\rho(k)||\psi_1(k_1)||\psi_{21}(k_{21})||\psi_{22}(k_{22})|$$

where $A_3 = |k_1|^{-\gamma-\alpha}|k_2(k-k_{21})(k_1+k_{21})|^{2\gamma-1}|k_{21}k_{22}|^{\gamma-\alpha-1}$ and use Cauchy-Schwartz on the double sum over k_1, k_{21} to bound

$$\Xi_2 \leq \sum_k |k|^{1-\gamma+\alpha}|\rho(k)| \left[\sum_{k_1,k_{21}} A_3^2\right]^{1/2} \left[\sum_{k_1,k_{21}} |\psi_1(k_1)|^2|\psi_{21}(k_{21})|^2|\psi_{22}(k_{22})|^2\right]^{1/2}$$

Now Lemma 22 below states that for $(\gamma,\alpha) \in \mathcal{D}$ we have $\sup_{k\in\mathbb{Z}_*} |k|^{1-\gamma+\alpha}[\sum_{k_1,k_{21}} A_3^2]^{1/2} \leq C$ so that

$$\Xi_2 \leq C \sum_k |\rho(k)| \left[\sum_{k_1,k_{21}} |\psi_1(k_1)|^2|\psi_{21}(k_{21})|^2|\psi_{22}(k_{22})|^2\right]^{1/2}.$$

Then using again Cauchy-Schwartz (on the k summation) we end up with

$$\Xi_2 \leq C|\rho|_0 \left[\sum_{k,k_1,k_{21}} |\psi_1(k_1)|^2|\psi_{21}(k_{21})|^2|\psi_{22}(k_{22})|^2\right]^{1/2} = C|\rho|_0|\psi_1|_0|\psi_{21}|_0|\psi_{22}|_0$$

where the last equality is obtained after a simple change of variables in the summations. This ends the proof of the statement. ■

It remains to prove some complementary results. We will use the notation introduced above in the proof of Lemma 5.

Lemma 21 *If $\gamma \in [0,1/2]$ and $h \neq 0$ we have*

$$|M| \leq C(A_1 + A_2)|t-s|^{2\gamma}$$

If $h = 0$

$$|M| \leq C|k_1|^{-1}|t-s|$$

Proof. Take two numbers $a, b \in \mathbb{R}$ and let $m = \int_s^t \int_s^\sigma e^{ia\sigma} e^{ib\sigma_1} d\sigma d\sigma_1$. An explicit integration gives

$$m = \frac{e^{i(a+b)s} - e^{i(a+b)t}}{b(a+b)} + \frac{e^{ibs+iat} - e^{i(a+b)s}}{ab}.$$

Then

$$|m| \leq \frac{1}{|b||a+b|} + \frac{1}{|a||b|} \tag{57}$$

moreover

$$|m| \leq \int_s^t \left| \int_s^\sigma e^{ib\sigma_1} d\sigma_1 \right| d\sigma \leq \frac{|t-s|}{|b|}$$

and symmetrically $|m| \leq |t-s||a|^{-1}$. These last bounds imply that $|m| \leq |t-s||ab|^{-1/2}$ and moreover imply the bound in the case $h = 0$. Otherwise, since $2\gamma \in [0,1]$, interpolating between this bound and eq. (57) we get

$$|m| \leq \frac{|t-s|^{2\gamma}}{|ab|^\gamma} \left(\frac{1}{|b||a+b|} + \frac{1}{|a||b|} \right)^{1-2\gamma} \leq \frac{|t-s|^{2\gamma}}{|b|^\gamma |a|^{1-\gamma} |a+b|^{1-2\gamma}} + \frac{|t-s|^{2\gamma}}{|a|^{1-\gamma}|b|^{1-\gamma}} \tag{58}$$

which, setting $a = 3kk_1k_2$ and $b = 3k_2k_{21}k_{22}$ implies the statement we had to prove. ∎

Lemma 22 *Let*

$$S_\varepsilon(k) = \sum_{k_1, k_{21}} |k_2|^{4\gamma-2+2\varepsilon} |k_1|^{-2\gamma-2\alpha+2\varepsilon} |(k-k_{21})(k_1+k_{21})|^{4\gamma-2} |k_{21}k_{22}|^{2\gamma-2\alpha+2\varepsilon-2}$$

Then if $(\gamma, \alpha) \in \mathcal{D}$, there exists $\varepsilon > 0$ such that

$$|k|^{2-2\gamma+2\alpha+\varepsilon} S_\varepsilon(k) \leq C$$

uniformly on $k \in \mathbb{Z}_*$.

Proof. We will prove the statement for $\varepsilon = 0$, the extension for a positive but small ε being a straightforward modification of the resulting argument.

First we bound the double sum by a planar integral. Define the integration region $\mathcal{R}(z,a)$ with $a > 0$, $z \in \mathbb{R}$ as the set of points $(x,y) \in \mathbb{R}^2$ such that

$$|x| > a, |y| > a, |x-z| > a, |y-z| > a, |x+y| > a, |x+y-z| > a$$

Then

$$S_0(k) \leq C \int_{\mathcal{R}(k,1)} dxdy |k-x|^{4\gamma-2} |x|^{-2\gamma-2\alpha} |(k-y)(x+y)|^{4\gamma-2} |y(k-x-y)|^{2\gamma-2\alpha-2}$$

which, by scaling away $|k|$ and by reflection symmetry, is equivalent to

$$S_0(k) \leq C|k|^{2(4\gamma-2)+3(2\gamma-2\alpha-2)+2} I(\mathcal{R}(1, |k|^{-1}))$$

with

$$I(\mathcal{A}) = \int_{\mathcal{A}} dxdy|1-x|^{4\gamma-2}|x|^{-2\gamma-2\alpha}|(1-y)(x+y)|^{4\gamma-2}|y(1-x-y)|^{2\gamma-2\alpha-2}.$$

To prove the original statement we must show that $|a|^{2(3-6\gamma+2\alpha)}I(\mathcal{R}(1,a)) \leq C$ uniformly in $a \to 0$.

We start by showing that $I(\mathcal{R}(1,a))$ is actually finite for any $a > 0$. Then we will find estimates for $I(a)$ as $a \to 0$.

Boundedness Note that, when $|x|+|y|$ is large we can bound the integrand of $I(\mathcal{R}(1,a))$ by

$$C|x|^{2\gamma-2\alpha-2}|y(x+y)|^{6\gamma-2\alpha-24}$$

Then since in $\mathcal{R}(1,a)$ we have $|x| > a$ we can bound this last integrand by $C|a|^{2\gamma-2\alpha-2}|y(x+y)|^{6\gamma-2\alpha-4}$. Using the fact that $6\gamma - 2\alpha - 4 < -1$ in $\mathcal{D}$, the integral $\int_{\mathcal{R}(1,a)} |y(x+y)|^{6\gamma-2\alpha-4}dxdy$ can be shown to be finite.

Asymptotics When $a \to 0$ the integral can blow up due to contributions from the sub-domain $\mathcal{R}'(a) = \mathcal{R}(1,a)\backslash\mathcal{R}(1,a')$ where $a' > 0$ is a small but fixed quantity. Denote with $B(z,r) \subset \mathbb{R}^2$ the ball centered in z and of radius r. The region $\mathcal{R}'(a)$ can be covered by the union of the seven sets

$$\mathcal{B}_i = \mathcal{R}'(a) \cap B(z_i, a'),\ i = A, \ldots, F\ , \qquad \mathcal{R}''(a) = \mathcal{R}'(a)\backslash\left[\cup_{i=A,\ldots,F}\mathcal{B}_i\right]$$

where z_i,$i = A, \ldots, F$ are the following points

$$z_A = (-1,1),\ z_B = (0,1),\ z_C = (1,1),\ z_D = (0,0),\ z_E = (1,0),\ z_F = (1,-1).$$

We will study the contributions to $I(\mathcal{R}'(a))$ due to $I(\mathcal{B}_i)$, $i = A, \ldots, F$. It is not difficult then to show that $I(\mathcal{R}''(a))$ gives smaller contributions to the $a \to 0$ limit.

Let us show how to handle $I(\mathcal{B}_A)$. First bound

$$I(\mathcal{B}_A) \leq C \int_{\mathcal{B}_A} |(1-y)(x+y)|^{4\gamma-2}dx\,dy$$

the other factors being bounded as $a \to 0$ for $(x,y) \in \mathcal{B}_A$. By a change of variables we can reduce this integral to the following form (eventually increasing the integration domain) $I(\mathcal{B}_A) \leq C\int_a^{a''} |xy|^{4\gamma-2}dxdy$ where $a'' > a$ is another fixed constant. Then we easily compute the factorized integral and finally bound the original expression with

$$I(\mathcal{B}_A) \leq Ca^{-2(1-4\gamma)_+} = Ca^{\zeta_A}$$

where $(\cdot)_+$ denotes the positive part. The constant C in this expression does not depend on a.

The treatment of all $I(\mathcal{B}_i)$ for $i = B, \ldots, F$ follows similar lines. To simplify the argument it is not difficult to see that $I(\mathcal{B}_i)$ for $i = A, C, F$ have the same asymptotic bound. The same is true for $I(\mathcal{B}_i)$ for $i = B, D$ where

$$I(\mathcal{B}_{B,D}) \leq C \int_{\mathcal{B}_B} |x|^{-2\gamma-2\alpha}|(1-y)|^{4\gamma-2}|x+y-1|^{2\gamma-2\alpha-2}dx\,dy$$

Moreover

$$I(\mathcal{B}_E) \leq C \int_{\mathcal{B}_E} |x-1|^{4\gamma-2}|y|^{2\gamma-2\alpha-2}|x+y-1|^{2\gamma-2\alpha-2}dx\,dy$$

and since $-2\gamma-2\alpha > 2\gamma-2\alpha-2$ for $\gamma \in [0,1/2)$ we have also $I(\mathcal{B}_{B,D}) \leq CI(\mathcal{B}_E)$. Since we already have established the behavior of $I(\mathcal{B}_A)$ it remains only to understand that of $I(\mathcal{B}_E)$. We have two possibilities. The first is to bound $|x+y-1|^{2\gamma-2\alpha-2} \leq |a|^{2\gamma-2\alpha-2}$ in $\mathcal{B}_E$, then

$$\begin{aligned} I(\mathcal{B}_E) &\leq C|a|^{2\gamma-2\alpha-2} \int_{\mathcal{B}_E} |x-1|^{4\gamma-2}|y|^{2\gamma-2\alpha-2}dx\,dy \\ &\leq C|a|^{2\gamma-2\alpha-2-(1-4\gamma)_+-(1-2\gamma+2\alpha)_+} = Ca^{\zeta_{E,1}} \end{aligned}$$

by reasoning as in the case of $I(\mathcal{B}_A)$ above. The second is to bound $|x-1|^{4\gamma-2} \leq |a|^{4\gamma-2}$ to obtain

$$\begin{aligned} I(\mathcal{B}_E) &\leq C|a|^{4\gamma-2} \int_{\mathcal{B}_E} |x+y-1|^{2\gamma-2\alpha-2}|y|^{2\gamma-2\alpha-2}dx\,dy \\ &\leq C|a|^{4\gamma-2-2(1-2\gamma+2\alpha)_+} = Ca^{\zeta_{E,2}} \end{aligned}$$

and let $\zeta_E = \max(\zeta_{E,1}, \zeta_{E,2})$.

In the same way it is not difficult to show that $I(\mathcal{R}''(a)) \leq C(|a|^{\zeta_A} + |a|^{\zeta_E})$. Putting all these estimates together we get $I(\mathcal{R}(1,a)) \leq C(|a|^{\zeta_A} + |a|^{\zeta_E})$. It remains to prove that

$$\zeta'_{A,E} = 2(3-6\gamma+2\alpha) + \zeta_{A,E} \geq 0 \tag{59}$$

for any admissible couple (γ,α). We proceed by considering three different regions $C_{1,2,3}$ of admissible couples:

- C_1 : $\gamma \in [0,1/4)$, $1-2\gamma+2\alpha < 0$. Here $\zeta_A = -2(1-4\gamma)$ and $\zeta_E = \zeta_{E,1} = 6\gamma-2\alpha-3$. Then $\zeta'_A = 2(2-2\gamma+2\alpha)$ and $\zeta'_E = 3-6\gamma+2\alpha$;
- C_2 : $\gamma \in [0,1/4)$, $1-2\gamma+2\alpha > 0$. Here $\zeta_A = -2(1-4\gamma)$ and $\zeta_E = \zeta_{E,1} = 8\gamma-4\alpha-4$. Here $\zeta'_A = 2(2-2\gamma+2\alpha)$ and $\zeta'_E = 2(1-2\gamma)$;
- C_3 : $\gamma \in [1/4,1/2)$, $1-2\gamma+2\alpha > 0$. Here $\zeta_A = 0$ and $\zeta_E = \zeta_{E,2} = 4\gamma-4\alpha-3$. Here $\zeta'_A = 2(3-6\gamma+2\alpha)$ and $\zeta'_E = 2(1-2\gamma)$.

We can check that in the three regions $C_{1,2,3}$ and for $(\gamma,\alpha) \in \mathcal{D}$ we always end up with $\zeta'_{A,E} > 0$. By interpolation between these three regions we prove the claim for couples in their convex hull which coincide with $\mathcal{D}$.

To treat the case of small positive ε we just retrace this argument and take ε small enough so that all relevant inequalities are still satisfied. ■

A.3 Proof of Lemma 12

Proof. Let us work out the details for X, first. Since $(\gamma,\alpha) \in \mathcal{D}$, by Lemma 1 there exists $\varepsilon > 0$ such that $X \in \mathcal{C}_2^\gamma \mathcal{L}((H^{\alpha-\varepsilon})^2, H^{\alpha+\varepsilon})$. By definition

$$|X_{ts}^{(N)}(\varphi_1,\varphi_2)|_\alpha = |P_N X_{ts}(P_N\varphi_1, P_n\varphi_2)|_\alpha$$

Then

$$|(X_{ts}^{(N)} - X_{ts})(\varphi_1, \varphi_2)|_\alpha \leq |\widetilde{P}_N X_{ts}(P_N\varphi_1, P_N\varphi_2)|_\alpha + |X_{ts}(\widetilde{P}_N\varphi_1, P_N\varphi_2)|_\alpha + |X_{ts}(\varphi_1, \widetilde{P}_N\varphi_2)|_\alpha \tag{60}$$

where $\widetilde{P}_N = \mathrm{Id} - P_N$ is the projector over modes with $|k| > N$. Now note that, for $\varphi \in H^\alpha$ we have $|\widetilde{P}_N\varphi|_{\alpha-\varepsilon} \leq N^{-\varepsilon}|\varphi|_\alpha$ so the three terms in eq. (60) can be bounded as follows. The first as

$$\begin{aligned}|\widetilde{P}_N X_{ts}(P_N\varphi_1, P_N\varphi_2)|_\alpha &\leq N^{-\varepsilon}|X_{ts}(P_N\varphi_1, P_N\varphi_2)|_{\alpha+\varepsilon} \\ &\leq CN^{-\varepsilon}|P_N\varphi_1|_{\alpha-\varepsilon}|P_N\varphi_2|_{\alpha-\varepsilon} \leq CN^{-\varepsilon}|\varphi_1|_\alpha|\varphi_2|_\alpha,\end{aligned}$$

the second as

$$|X_{ts}(\widetilde{P}_N\varphi_1, P_N\varphi_2)|_\alpha \leq C|\widetilde{P}_N\varphi_1|_{\alpha-\varepsilon}|P_N\varphi_2|_{\alpha-\varepsilon} \leq CN^{-\varepsilon}|\varphi_1|_\alpha|\varphi_2|_\alpha,$$

and a similar bound for the third. Then

$$|(X_{ts}^{(N)} - X_{ts})(\varphi_1, \varphi_2)|_\alpha \leq CN^{-\varepsilon}|\varphi_1|_\alpha|\varphi_2|_\alpha \tag{61}$$

which implies the claimed convergence.

Now consider the convergence of $\hat{X}^{(N),2}$ to $\hat{X}^{(2)}$. We want to show that

$$|(\hat{X}_{ts}^{(N),2} - \hat{X}_{ts}^2)(\varphi_1, \varphi_2, \varphi_3)|_\alpha \leq CN^{-\varepsilon}|\varphi_1|_\alpha|\varphi_2|_\alpha|\varphi_3|_\alpha \tag{62}$$

for some $\varepsilon > 0$, this will imply the claim. Define a multiplier Θ as

$$\Theta = \chi_N(k)\chi_N(k_1)\chi_N(k_2)\chi_N(k_{21})\chi_N(k_{22}) - 1$$

where $\chi_N(k) = 1_{|k|\leq N}$ is the kernel in the Fourier basis of the operator P_N. By duality we need to bound the form

$$\sum_{k,k_1,k_{21}} |M||\Theta||k|^\alpha|k_1|^{-\alpha}|k_{21}|^{-\alpha}|k_{22}|^{-\alpha}|\rho(k)||\psi_1(k_1)||\psi_{21}(k_{21})||\psi_{22}(k_{22})| \tag{63}$$

with $CN^{-\varepsilon}|t-s|^{2\gamma}|\rho|_0|\psi_1|_0|\psi_{21}|_0|\psi_{22}|_0$. Recall that the multiplier M has been defined in eq. (56). The Θ multiplier can be rewritten as a telescopic sum

$$\begin{aligned}\Theta = &[\chi_N(k) - 1]\chi_N(k_1)\chi_N(k_2)\chi_N(k_{21})\chi_N(k_{22}) + [\chi_N(k_1) - 1]\chi_N(k_2)\chi_N(k_{21})\chi_N(k_{22}) \\ &+ [\chi_N(k_2) - 1]\chi_N(k_{21})\chi_N(k_{22}) + [\chi_N(k_{21}) - 1]\chi_N(k_{22}) + [\chi_N(k_{22}) - 1]\end{aligned}$$

and then can be bounded as

$$|\Theta| \leq |\chi_N(k) - 1| + |\chi_N(k_1) - 1| + |\chi_N(k_2) - 1| + |\chi_N(k_{21}) - 1| + |\chi_N(k_{22}) - 1|$$

Moreover since

$$|\chi_N(k) - 1| \leq |\chi_N(k) - 1|\frac{|k|^\varepsilon}{N^\varepsilon} \leq N^{-\varepsilon}|k|^\varepsilon$$

for any $\varepsilon \geq 0$ we have that

$$|\Theta| \leq N^{-\varepsilon}(|k|^\varepsilon + |k_1|^\varepsilon + |k_2|^\varepsilon + |k_{21}|^\varepsilon + |k_{22}|^\varepsilon) \leq N^{-\varepsilon}|kk_1k_2k_{21}k_{22}|^\varepsilon$$

Then to obtain the required N dependence of the bound for the form (63) it is enough to bound the new N-independent form

$$\sum_{k,k_1,k_{21}} |M||k|^{\alpha+\varepsilon}|k_1|^{-\alpha+\varepsilon}|k_2|^{\varepsilon}|k_{21}|^{-\alpha+\varepsilon}|k_{22}|^{-\alpha+\varepsilon}|\rho(k)||\psi_1(k_1)||\psi_{21}(k_{21})||\psi_{22}(k_{22})| \tag{64}$$

with a quantity like $C|t-s|^{2\gamma}|\rho|_0|\psi_1|_0|\psi_{21}|_0|\psi_{22}|_0$. By retracing the proof of Lemma 5 and using the $\varepsilon > 0$ version of Lemma 22 we can prove that such bound holds for all admissible couples (γ, α).

This completes the proof of the statement.

■

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